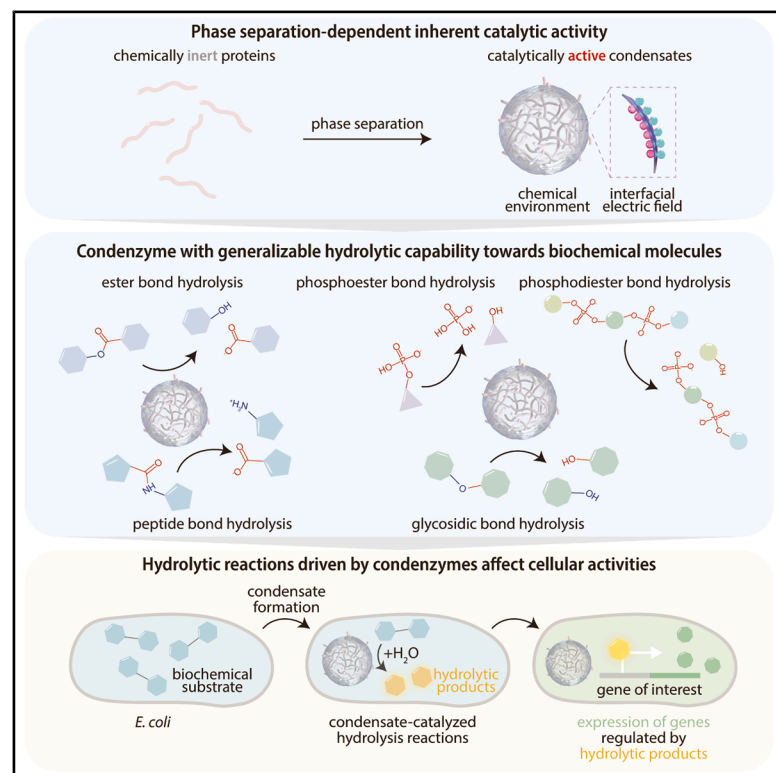


Biomolecular condensates can function as inherent catalysts

Graphical abstract



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In brief

Chen, Guo, Farag, Qian et al. present that biomolecular condensates formed by proteins lacking catalytic activity can mediate diverse hydrolysis reactions, highlighting the importance of condensate electrochemical environments in regulating condensate biochemistry.

Highlights

- Biomolecular condensates exhibit inherent catalytic activity
- Phase separation-dependent activation of catalytic function
- Condensate electrochemical microenvironments tune catalytic performance
- Condenzymes reprogram cellular signaling



Article

Biomolecular condensates can function as inherent catalysts

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SUMMARY

Biomolecular condensates possess distinct solvent environments. This has been postulated to affect biochemical activities. Here, we report the discovery of inherent catalytic functions of condensates formed by intrinsically disordered proteins. The proteins themselves lack any catalytic activities. The catalytic functions of condensates emerge as a consequence of phase separation-dependent mesoscale electrochemical microenvironments. Condensates can mediate the hydrolysis of diverse types of biologically relevant substrates. Through sequence design that enriches selective residues at condensate interfaces, interfacial electrochemical properties, such as interfacial electric field and water activity, are shown to control catalytic behaviors. Synthetic condensates in live cells exhibit hydrolytic activity, reshaping transcription profiles and activating gene circuits that depend on hydrolytic products. Together, these findings support a model that biomolecular condensates function as “condenzymes,” contributing emergent biochemical functions in cells.

INTRODUCTION

Biomolecular condensates, formed through biomacromolecular condensation, concentrate condensate-specific biomolecules to regulate diverse cellular functions¹ such as stress responses and ribosomal biogenesis.^{2–7} Condensation, which involves spontaneous phase separation of the solution system as a whole, results in the coexistence of two or more phases.^{8–16} The differences in solvent and biomolecular compositions between the condensate interiors and coexisting dilute phases result in distinct interphase physicochemical properties.^{8,17–19} These come about due to the establishment of phase equilibria whereby the chemical potentials of each of the species are equalized, thus giving rise to the possibility of asymmetric partitioning of these species across coexisting dense and dilute phases.¹⁷ The redistribution of solvent or asymmetry in net charge both lead to differential partition of solution ions across condensate interiors and coexisting dilute phases, which establish the interphase electric potentials.^{17,20–23} These interphase potentials are manifest as pH gradients sustained by co-existing phases.^{16,18,24} Lastly, at condensate interfaces where there is a sharp transition of molecular density,²⁵ electric potential gradi-

ents manifest as electric double layers,^{8,26–28} which establish an interfacial electric field.^{8,29,30}

The presence of distinct interphase and interfacial physicochemical properties, including asymmetries of ionic and hydration properties across coexisting phases and the presence of interfacial electric fields,^{8,13–15,25,26,30} suggests that distinct functionalities can be enabled by the chemical environments and electrochemical properties within condensates and their interfaces.^{8,31–33} This hypothesis derives from analogies with observations in colloid chemistry and micron-sized water droplets.^{34–38} A chemically active emulsion would imply that condensate functions are not determined solely by their capacity to store and compartmentalize biomolecules or organize biochemical reactions.^{39,40} They also have the potential to contribute as drivers of biochemical reactions through their distinct inherent physicochemical properties.²³

The growing inventory of distinct environmental features created within condensates and their interfaces inspired us to draw parallels with proteins enzymes,^{23,33} wherein the catalytic activities can be attributed to the local environments within the enzyme active sites.^{41,42} This includes the distinctive hydration environment and electric fields created by structural alignments



of functional residues.^{41–44} We hypothesized that the physicochemical properties within condensate interiors and interfaces might share similarities to those within the catalytic sites of enzymes. Therefore, while enzymes are nanoscale catalysts, we conjectured that condensates have the potential to function as mesoscale catalysts. Accordingly, we tested whether the physicochemical features of condensates can enable spontaneous catalytic functions.

In this study, we evaluate the capability of condensates to catalyze hydrolysis reactions both in vitro and in living cells. The underlying driving forces that mediate catalysis were studied using density functional theory (DFT) and examined via the electronic Stark effect. Using *de novo* designs of IDPs that were guided by coarse-grained simulations, we uncovered the contribution of the interfacial electric fields to the kinetics of catalytic reactions driven by condensates. Using stimulated Raman spectroscopy, we mapped the local water hydrogen-bonding network in condensates and found that the interior and the interface of condensates possess distinct water activity, which is sequence dependent. By applying different chemical assays across multiple condensate types, we show that hydrolysis is a generalizable inherent function of condensates, while the hydrolytic activity and selectivity are governed by condensate composition and substrate chemistry. We further employed cellular lysate assays, a knockout cell strain, and synthetic gene circuits to demonstrate that the catalytic functions of condensates are capable of modulating hydrolysis reactions and downstream pathways in living cells. Overall, we find that condensates formed by catalytically inactive IDPs can drive the catalysis of biochemical reactions through their electrochemical properties. We name such condensates as “condenzymes,” biomolecular condensates with the inherent ability to catalyze chemical reactions.

RESULTS

Hydrolysis reactions can be catalyzed by condensates

We first employed resilin-like polypeptides (RLPs),^{45,46} whose chemical environments and interfacial electrochemical properties have been characterized,^{8,19,26} to form condensates for evaluating whether condensates can drive hydrolysis reactions. Hydrolysis is a crucial biochemical process in live cells. For example, in the case of ATP hydrolysis, ATPases mediate the binding of ATP within the active site and cleavage of a bond by a water molecule,⁴⁷ enabling energy transduction and cellular signaling.^{48,49} Some hydrolysis reactions favor an alkaline environment,⁵⁰ and this matches the dense phase environments of RLP condensates^{8,19,26} (Figure 1A). Accordingly, we first explored the effects of condensates formed by the RLP_{SKGP} protein (amino acid sequence: Ser-Lys-Gly-Pro-[Gly-Arg-Gly-Asp-Ser-Pro-Tyr-Ser]₂₀-Gly-Tyr) on ester hydrolysis. Phase transitions of RLP_{SKGP} were triggered by reducing the ionic activity of the solution (Figures S1A and S1B). We applied a chromogenic assay using *para*-nitrophenyl acetate (pNPA), which is a widely used substrate for the evaluation of esterase activity.⁵¹ Hydrolysis of pNPA decomposes the substrate to acetate and *para*-nitrophenol (pNP) (Figure 1B), and the product, pNP, can be directly monitored by ultraviolet-visible (UV-vis) absorption

spectroscopy at a wavelength of 410 nm, which is the absorption wavelength of phenol. We first verified the linear correlation between pNP concentration and absorbance at 410 nm (Figure S1C). Subsequently, we evaluated the pNPA hydrolysis reaction using an absorption scan under three different solution conditions: solution with condensates ($C_{\text{RLP}} > C_{\text{sat}}$), solution without condensates ($C_{\text{RLP}} < C_{\text{sat}}$), and the buffer solution without protein. Here, C_{RLP} and C_{sat} are the bulk and saturation concentrations of RLP, respectively. We confirmed the formation or the absence of condensates using confocal differential interference contrast microscopy (Figure 1C). In solutions with condensates, the colorless pNPA turned yellow after 120 min of incubation at room temperature (Figure 1C). This was evident from the appearance of a product absorption band at 410 nm (Figure 1D). In the solution without condensates and in the buffer solution, the signal was markedly lower than in the solution containing condensates. The weak signals in condensate-free or buffer solutions derive from the spontaneous hydrolysis of pNPA. The same trend was observed in time-dependent reaction tracking experiments (Figure 1E). Overall, our results suggest that the presence of condensates enhances the hydrolysis of pNPA.

To test if the reaction products are concordant with hydrolysis of pNPA, we used linear ion mass spectroscopy to analyze the supernatant of the condensate solution with pNPA after 120 min of incubation at room temperature. Distinct peaks at m/z ratios of 138, 77, and 59 were observed (Figure 1F). These correspond to *p*-nitrophenol, acetate, and acetate hydrate, confirming the esterolysis of pNPA in the solution with condensates. To evaluate whether the catalytic reaction is governed by condensate formation, we applied the pNPA assay to solutions containing different concentrations of RLP_{SKGP} and quantified the initial velocity (V_0). The bulk concentration of RLP_{SKGP} is not linearly correlated with V_0 . Instead, we find that once the concentration of RLP_{SKGP} crosses its C_{sat} , V_0 increases exponentially (Figure 1G). This observation suggests that the catalytic function requires the formation of condensates.

To further confirm that the catalytic behavior is not protein concentration-dependent or due to contaminants from protein purification, we prepared RLP_{SKGP} solution (0–15 μM) in 378 mM NaCl buffer, a high-salt condition that abolishes condensate formation. Through this range, we observed no catalytic activity. The C_{sat} of RLP_{SKGP} under 150 mM NaCl is approximately 10 μM . These observations collectively confirm that the observed catalytic activity arises from the condensate formation rather than from the solution composition (Figures S1D and S1E).

To shed light on the mechanism by which condensates influence pNPA hydrolysis, we asked if the RLP_{SKGP} condensate acts as a catalyst, as opposed to being a reactant. As a catalyst, the material of the condensates should not be produced or destroyed by the pNPA decomposition reaction.⁵² We quantified the dense phase concentration using a sedimentation assay and compared the physical forms of the dense phase using confocal imaging. Across samples, with and without pNPA after 120 min of incubation at room temperature (Figure S1F), we did not observe significant differences in the dense phase concentrations or areas of the condensates (Figures S1G and S1H). These observations suggest that the reaction does not consume the components of the condensates.

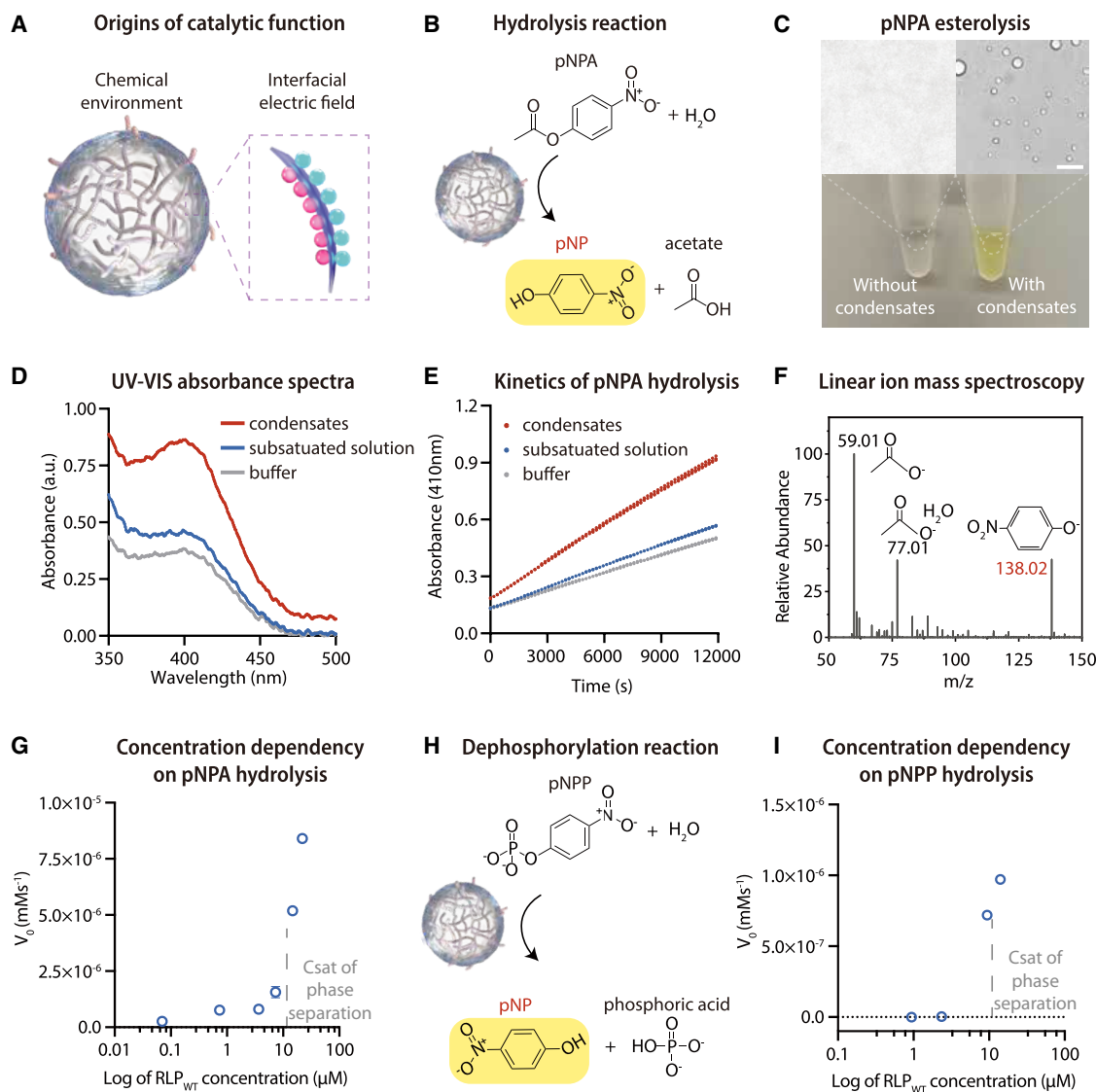


Figure 1. Biomolecular condensates can function as catalysts for hydrolysis reactions

(A) Biomolecular condensates are defined by a chemical environment in the dense phase and an electric field at their interface.

(B) The hydrolysis reaction of *p*-nitrophenol acetate (pNPA) results in the formation of *p*-nitrophenol (a yellow product) and acetate.

(C) Image visualization of the catalytic reaction with a change of color in the solution with condensates. The samples containing 0.5 mM pNPA with or without condensates were incubated at room temperature for 120 min. The two images on the top are representative images displaying solution with or without condensates. Scale bar, 10 μm.

(D) UV-vis absorbance spectrums of (1) the solution containing condensates (15 μM RLP_{SKGP}) and pNPA, (2) the subsaturated RLP_{SKGP} solution (3 μM) and pNPA, and (3) the buffer solution containing pNPA. The final concentration of pNPA was 0.5 mM.

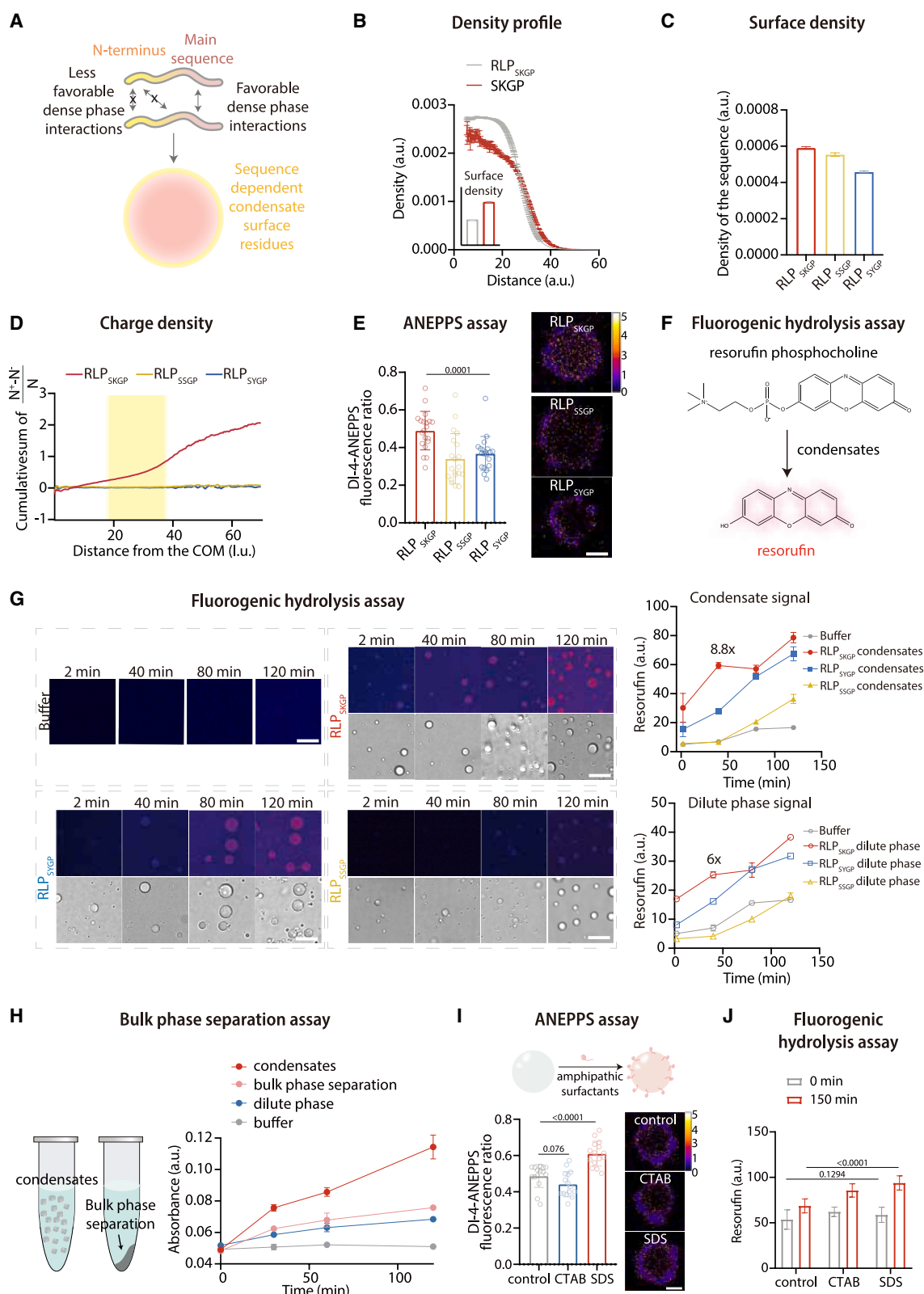
(E) Time-dependent absorbance tracking of pNPA hydrolysis with the same experimental setup as in (D). *N* = 4 independent experiments. Each data point represents mean ± SE.

(F) A sample spectrum of linear ion mass spectroscopy analysis of the hydrolysis of pNPA by RLP_{SKGP} condensates for 120 min at room temperature. The same analysis was performed three times with different samples, and the results were consistent.

(G) Protein concentration-dependent initial rate (*V*₀) of the pNPA catalytic reaction. The protein concentration at which a sharp transition of the initial rate happened aligned with the saturation concentration of the RLP_{SKGP}. Each data point represents mean ± SE.

(H) The hydrolysis reaction of *p*-nitrophenol phosphate (pNPP) results in the formation of *p*-nitrophenol (a yellow product) and phosphoric acid.

(I) Protein concentration-dependent initial rate (*V*₀) of the pNPP catalytic reaction. *N* = 3 independent experiments. Each data point represents mean ± SE.



(legend on next page)

Condensates can mediate hydrolysis of substrates with distinct chemical groups

To explore the ability of condensates to catalyze other hydrolysis reactions, we first investigated whether the RLP_{SKGP} condensates could mediate the hydrolysis of *p*-nitrophenyl phosphate (pNPP) (Figure 1H), which is a dephosphorylation process. Interestingly, we found that RLP_{SKGP} condensates also catalyze dephosphorylation in a phase separation-dependent manner (Figures 1I and S1I). We next employed compounds with distinct chemical groups next to the ester bond as substrates (Figure S1J), including *p*-nitrophenyl butyrate (pNPB), *p*-nitrophenyl trimethylacetate (pNP-TMA), and *p*-nitrophenyl octanoate (pNPO). We found that at the same substrate concentration and reaction condition, the rate of hydrolysis reactions differed significantly between different substrates (Figure S1J). This observation suggests that the substrate affinity for the condensate microenvironment, dictated by substrate chemistry, might contribute to catalytic specificity. For substrates that could be hydrolyzed efficiently by condensates, we found that the V_0 of the hydrolysis of all the substrates increases exponentially once the RLP_{SKGP} concentration crosses the c_{sat} (Figure S1K), confirming that the hydrolytic behavior is a phase separation-dependent feature. These findings suggest that (1) condensates can drive different hydrolysis reactions and (2) the coupling of the chemical properties of the substrate and the condensate environment can be critical to the catalytic performance of condensates. This lends credence to the idea that the chemical features of condensates provide a favorable environment for enhancing the rates of reactions, similar to the functioning mechanism of an enzyme.⁴²

Interfacial electric fields help lower the free-energy barrier for hydrolysis reactions

The fact that condensates can catalyze different chemical reactions implies that the catalytic activities of condensates are influenced by the electrochemical features defined by the environments of condensate solutions.⁸ We hypothesized that the

interfacial electric field might be one of the features of condensates that can result in lowering the activation free energy of reactions.^{53,54} To explore the physical basis of this possibility, we studied the influence of external electric field strengths on the rates of pNPA hydrolysis using density functional theory (DFT).⁵⁵ We evaluated the difference in dipole moment between the initial substrate and the transition state of the pNPA to quantify the field-induced change of the reaction barrier (see STAR Methods for details). Under the influence of an electric field strength of $\sim 10^7$ V/cm (a typical value for the electrostatic environment in the enzymatic active site⁵⁶), the free-energy barrier ($\Delta\Delta G^\ddagger$) decreases from 9.26 to 7.73 kcal/mol at the intermediate state (Figure S1L). This lowering of the barrier should correspond to an approximate 13-fold increase in the rate of pNPA hydrolysis, which is comparable to the observations in our experiments. This result suggests that the interfacial electric fields of condensates are likely to play an important role in catalyzing chemical reactions.

Programming IDP sequence to modulate the interfacial electric fields of condensates

To understand how the interfacial electric fields contribute to catalytic reactions, we explored whether we could design IDP sequences to modulate the surface charge density of condensates without changing dense phase environments. We hypothesized that the surface charge density of condensates, which should be encoded in the sequence composition, would define the strength of the interfacial electric field,^{57,58} thereby modulating the catalytic behaviors. Similar to colloidal particles in a solution containing electrolytes,^{59,60} in an ideal condition, the first layer of ions within the Stern layer,⁶¹ which is governed by the surface charge of the colloid particle, will organize the assembly of the second layer of ions that set up the electric double layer. As such, the amino acid composition on the condensate surface should modulate the strength of the interfacial electric field.

Figure 2. Decoupling the contribution of the dense phase chemical environment and the interfacial electric field to the catalytic function of condensates

- (A) Schematic showing that regions of an IDP can have different intrinsic preferences for localizing to the dense phase versus localizing to the interfaces of condensates.
- (B) Radial density plots of the N-terminal region (first four residues) and full-length protein from simulations of the full-length RLP_{SKGP}. Each data point represents mean $\pm$ SE. Inset shows the comparison of the density of the N-terminal sequence and the full-length sequence at the interface of the condensates based on the radial density plots. Bar graph shows the mean $\pm$ SE.
- (C) Data from the radial density plots in (B) were used to compute the interfacial densities of the N-terminal motifs in different condensates.
- (D) Comparison of the cumulative density of net charge computed along the radially resolved coordinate from the center of mass of each condensate for each system. Bar graph shows the mean $\pm$ SE.
- (E) Evaluation of the interfacial electric field of different sequences (RLP_{SKGP}, RLP_{SYGP}, and RLP_{SSGP}, all with 15 μ M concentration) using ratiometric DI-4-ANEPPS fluorescence assay. The insets show the sample ratiometric images of condensates (at similar size) with DI-4-ANEPPS. Scale bar, 1 μ m. Statistical significance was analyzed by an unpaired *t* test.
- (F) Schematic showing how fluorogenic hydrolysis reaction of resorufin phosphocholine generates fluorescent resorufin.
- (G) Comparison of the resorufin signal of the fluorogenic assay using different condensates formed by RLPs with different RLP sequences (all with 15 μ M concentration). All condensates were formed in 150 mM NaCl, 50 mM Tris, and pH 7.5. The excitation was set at 550 nm with a WLL laser, and the emission detector was set at 570–600 nm on a HYD detector. The gray line represents the spontaneous hydrolysis of resorufin phosphocholine in the same buffer solution. Scale bar is 5 μ m. Each data point represents mean $\pm$ SE.
- (H) Schematic illustration of the experimental design for creating bulk-phase separation by spinning down the condensates to reduce the effective surface areas and number of interfaces. *N* = 3 independent experiments. Each data point represents mean $\pm$ SD.
- (I) ANEPPS assay analysis of the interfacial electric field of RLP_{SKGP} condensates after being treated with surfactants. Scale bar, 1 μ m.
- (J) Fluorogenic hydrolysis assay based on resorufin phosphocholine analysis of the change of hydrolytic capability of RLP_{SKGP} condensates treated with surfactants. Statistical significance was analyzed by an unpaired *t* test.

We hypothesized that if certain regions of IDPs preferentially accumulate at the surface of the condensate, then we should be able to tune the surface charges of condensates.⁶² This would allow us to alter the interfacial ion organization (Figure 2A). To this end, we first employed a coarse-grained simulation approach using LaSSI,⁶³ which is a Monte-Carlo simulation engine that has been used to model phase behaviors of low-complexity sequences and the internal organization of condensates.^{25,64} The simulations helped us explore if there is a defined region of RLP_{SKGP} that has a higher preference of localizing to condensate surface. The repetitive nature of the RLP sequence mediates strong homotypic interactions. We focused our attention on the N terminus (Ser-Lys-Gly-Pro), which lacks sticker residues^{65,66} and is distinct from the repetitive main sequence. We hypothesized that the N terminus should interact weakly with repetitive elements within the dense phase and that the charged Lys residue in the N terminus might drive a preference for the N terminus to localize to the condensate surface.

We performed simulations using the full-length protein and analyzed the resulting radial density profiles. This analysis focused on the localization preferences of the N terminus versus the rest of the sequence. As expected, the N terminus shows a decreased density in the dense phase (Figure 2B) and an increased density at the interface when compared with either the C terminus or the full protein. This observation suggests that the specific sequence of the N terminus (Ser-Lys-Gly-Pro) may promote its localization to the interface.

Next, we made variants containing different N-terminal sequences by replacing the charged residue, Lys, in the N terminus sequence with a Ser or Tyr. This sets up a hierarchy whereby the wild-type (RLP_{SKGP}) N terminus, which contains a Lys residue, would have the weakest interaction with the main repeat, followed by the N terminus with a Ser residue (RLP_{SSGP}), and the N terminus with a Tyr residue (RLP_{SYGP}), which should have the strongest interaction with the main repeat. Indeed, we observed that the density of the N-terminal sequence motif in the interface was significantly diminished for the variants when compared with the RLP_{SKGP} sequence, with the N terminus containing a Tyr showing the lowest preference for localization to the interface (Figure 2C). This finding confirms that the relative interaction strengths between distinct sequence regions can affect the spatial preferences of each region within a condensate.⁶⁴ This also provides a fundamental ability to program the surface of condensates with minimal disruption on the interactions within dense or dilute phases, thereby allowing us to isolate contributions from the interfacial electric field to catalytic behaviors.

To understand how sequence variations contribute to charge asymmetry, which is the key to set up a potential gradient, the radial density profiles were computed for positively charged and negatively charged residues in the system to analyze the net charge across the radial coordinate from the centers of mass of different condensates in the simulations. The cumulative sum of the net charge radial density profile compared with the radial distribution of all beads in the system shows whether there is an accumulation of positively charged or negatively charged beads at specific regions, within the condensate, in the interfacial region (shaded), and in the dilute phase (Figure 2D). Indeed, for the RLP_{SKGP} sequence, the concavity of the profile in the

interface shows accumulation of positively charged residues. For RLP_{SSGP} and RLP_{SYGP}, we did not observe any significant changes in the accumulation of net charge across the radial distances from the center of mass of the droplet. It is important to note that the surface charge properties of condensates arise collectively from the distributions of both proteins and ions, as well as their inter- and intramolecular interactions at the soft interface, features that cannot be fully captured by our simulations. Nevertheless, these results imply that the RLP_{SKGP} sequence with the highest charged contents among the sequences has the greatest capacity to organize a potential gradient.

We tested the predictions from simulations using experiments that investigated the effects of distinct sequences on the interfacial electric field and the catalytic behaviors of the resulting condensates. We hypothesized that mutations of the charged residue at the soft surface of condensates might alter ion alignment and charge density,^{67,68} thereby modulating the strength of the interfacial electric field and catalytic capabilities. To ensure that the modification of a single-amino acid residue at the N terminus does not affect the internal properties of condensates, we first measured the dense phase concentrations of these condensates and did not observe significant differences (Figure S2A). This suggests that the homotypic interactions between the repetitive sequences are the main stabilizers of the dense phase. Next, we probed the pH of these condensates using a previously established method^{8,24} and found that the N terminus sequence does not significantly affect the dense phase pH (Figure S2B), confirming that the main repetitive sequences determine the chemical environments of the dense phase.

To investigate how the interfacial properties of condensates change due to single mutations in the N terminus, we used a ratiometric, electronic Stark effect probe, DI-4-ANEPPS, which undergoes a structural change and fluoresces upon experiencing a local electric field.^{8,69,70} Compared with typical zeta potential measurements, which provide indirect estimates of interfacial electric fields and necessitate theoretical corrections for the fluctuating nature of soft surfaces relative to typical ideal hard surfaces,^{71,72} the application of Stark effect probes enables direct characterization of the interfacial field. Further, the ratiometric feature of ANEPPS allows us to evaluate the electric field strength without considering the concentration-dependent effect of the dye.⁶⁹ We compared condensates of similar sizes and found a noticeable difference in the interfacial signal between the RLP_{SKGP}, RLP_{SSGP}, and RLP_{SYGP} sequences. The interfacial field is the strongest for RLP_{SKGP}, followed by RLP_{SYGP}, and weakest for RLP_{SSGP}, with an over 1.45-fold lower ANEPPS signal ratio generated by RLP_{SSGP} (Figure 2E). These observations align with the simulation results and suggest that the interfacial alignment of ions, which can be modulated by surface charge dictated by the protein sequence, can modulate the strength of the interfacial electric field.

Interface-dependent catalytic activity of condensates

Sequences that encode similar dense phase chemical environments while featuring distinct interfacial features enable the evaluation of contributions of interfacial electric fields to the catalytic functions of condensates. To study this at a single-condensate

level, we designed a fluorogenic assay using resorufin phosphocholine as the substrate. Hydrolysis of this substrate removes the phosphocholine and yields a fluorescent resorufin (Figure 2F). We first confirmed that the RLP proteins themselves are incapable of promoting the hydrolysis of resorufin phosphocholine (Figure S2C). We next tracked the catalytic behaviors of condensates formed by different sequences as a function of time (Figure 2G). Specifically, we compared the resorufin signal in different condensates of similar size ($r = 1\text{--}3\ \mu\text{m}$). The resorufin signal of condensates formed by RLP_{SKGP} was significantly higher than that of condensates formed by the mutants RLP_{SSGP} and RLP_{SYGP}. The most significant difference was recognized at 40 min with an 8.8-fold catalytic product difference between the condensates of RLP_{SKGP} and the condensates formed by RLP_{SSGP} (Figure 2G). We also observed that the hydrolytic activity of condensates led to increases of resorufin signal in the dilute phase through time (Figure 2G), suggesting that the increase of signal in the dense phase is not a partitioning effect, which is further confirmed with a control experiment based on time-dependent resorufin partitioning analysis using different types of RLP protein condensates (Figure S2D). Taken together, the trends in resorufin signals, which reflect the catalytic activity of different condensates, correlate with their interfacial potentials.

Next, we assessed whether the dilute phase environment is critical to the catalytic behavior of condensates. To this end, we modulated the bulk-phase environmental pH of solutions containing RLP condensates. This experiment was motivated by previous discoveries showing that the dense phase pH of RLP_{SKGP} condensates does not shift significantly upon a change in the bulk-phase pH.⁸ With the fluorogenic assay, we found that compared with condensates prepared in solutions that are more basic, a considerably higher catalytic activity was observed from RLP_{SKGP} condensates in solutions that are more acidic (Figures S2E and S2F), even though a basic bulk environment is favorable for the hydrolysis of resorufin phosphocholine (Figure S2G). We also confirmed that the different pH conditions do not affect the partitioning of resorufin (Figure S2H). We reasoned that the decrease of the bulk-phase pH, which mainly alters the dilute phase pH, sets up a higher pH gradient between the dilute and the dense phases (Figure S2I), thus resulting in a stronger interfacial electric field as observed previously.⁸ Taken together with the data that RLP condensates with distinct sequences or under distinct environments, our observations suggest that the interfacial electrochemical properties that are set up by electrochemical-environment differences between coexisting dilute and dense phases are critical to the catalytic behaviors of condensates.

To investigate the specific contributions of the interfaces of condensates, we designed two different assays to isolate the contribution of the interfaces of individual condensates to catalytic functions. We used bulk-phase separation to generate a single interface that delineates the macroscopic dilute and dense phases (Figure 2H). We implemented the pNPP assay to study the contributions of individual phases to the hydrolysis reaction in bulk solutions (see STAR Methods for details). Compared with the condensate solution, in which many interfaces exist, the solution that was separated into coexisting mac-

rophases showed 2.44-fold decreases of the catalytic behavior (Figure S2J). By extracting the dilute phase, we found a slightly higher level of catalytic activity compared with that of the buffer solution. This might be attributable to the presence of nanoscale clusters in subsaturated solutions.^{73–75} To further isolate interfacial effects, we deployed the anionic and cationic amphipathic surfactants, sodium dodecylsulfate (SDS) and cetyltrimethylammonium bromide (CTAB), to modulate the surface charge conditions of the condensates (Figure 2I). This should affect the alignment and distributions of surface ions,⁷⁶ thereby affecting the interfacial field of the condensates. We verified the change of interfacial field of individual condensates upon the addition of surfactants with the ANEPPS assay (Figure 2I). We then applied the resorufin phosphocholine-based fluorogenic hydrolysis assay and compared the change of the generated resorufin signal of these condensates with similar size during the same amount of reaction time. We found that compared with CTAB, condensates supplemented with SDS, which exhibited strengthened interfacial electric fields, significantly enhanced hydrolytic capability (Figure 2J). These results collectively confirm that the interfacial electric field of condensates is crucial to their catalytic performance.

Interface-enhanced water activity modulates hydrolytic activity of condensates

A key environmental property that could be modulated by the existence of a liquid-liquid interface and its associated electric field is the structural organization of water molecules,⁷⁷ which encodes the activity of water molecules.⁷⁸ Water molecules directly participate in hydrolysis reactions. Hence, water activity should directly affect hydrolysis reactions.⁷⁹ The abundance of free water molecules—those not hydrogen-bonded to each other but instead available to interact with solutes or substrates—has been correlated with the rates of catalytic reactions.^{80,81} Water molecules can also contribute to the establishment of interfacial electric field. As a result, in addition to the surface electrostatics that define the interfacial electric fields as described by the Gouy-Chapman-Stern formalism,⁸² the total field generated by the two-phase system can also be influenced by the solvent dipoles that are mediated by solvent-solute interactions.⁸³ An example of this is the Onsager reaction field at an electrochemical interface.⁸⁴ Given these considerations, we reasoned that as the polarity of water molecules and the interfacial electrostatic field are always coupled together, one might affect the other or vice versa. Therefore, different water activities might exist in a spatially heterogeneous manner, which might account for the differences in hydrolytic capacity between condensates.

To test for the possibility of spatially varying water activities, we employed a recently developed stimulated Raman scattering (SRS)^{85–88} based technique to study the hydrogen-bonding environment by imaging the local water status in a label-free manner⁷⁸ (Figure 3A). Harnessing the hyperspectral chemical imaging capability and submicron resolution of SRS microscopy, we mapped the authentic water vibrational signatures within the O–H stretching window for the three distinct RLP condensates (Figure 3A). The Raman signature of water O–H stretching is particularly sensitive to the hydrogen-bonding environment, in which three distinct types of water molecules can be

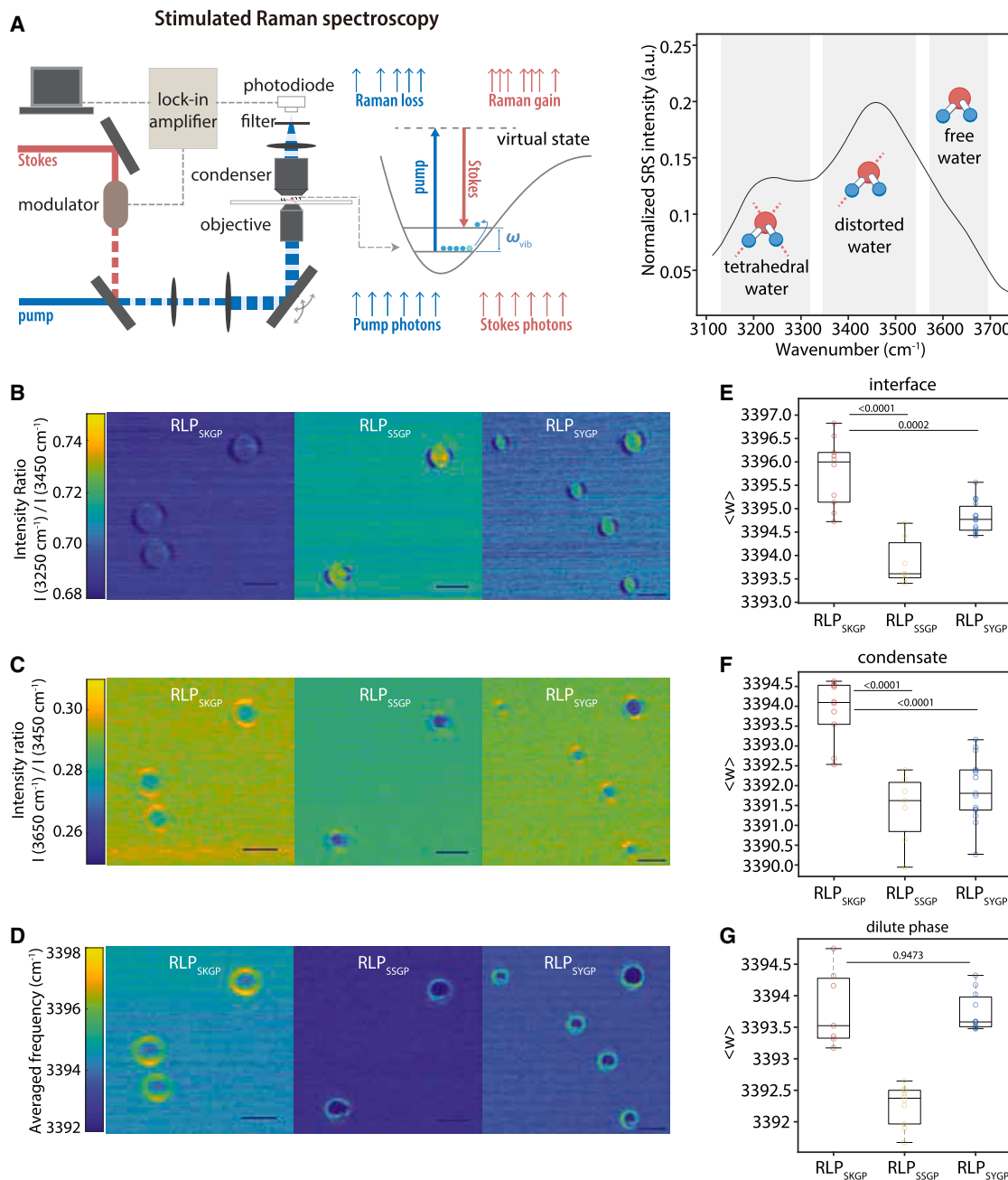


Figure 3. Characterization of water environments using hyperspectral stimulated Raman spectroscopy

(A) SRS microscopy illustration. The pump laser beam and modulated Stokes laser beam are combined and focused onto the sample. When the energy difference between the pump and the Stokes laser matches the vibrational energy (ω_{vib}) of the chemical bond of interest, the vibrational transition is accompanied by stimulated Raman loss (SRL) on the pump beam and stimulated Raman gain (SRG) on the Stokes beam. SRL on the pump beam is detected by the photodiode and extracted by the lock-in amplifier for SRS imaging.

(B) Hyperspectral SRS mapping of water heterogeneity in condensates with ratiometric SRS images ($I(3,250 \text{ cm}^{-1})/I(3,450 \text{ cm}^{-1})$) shows the relative abundance of tetrahedral water to distorted water. All RLP variants were used at a concentration of 15 μM . Scale bar, 3 μm .

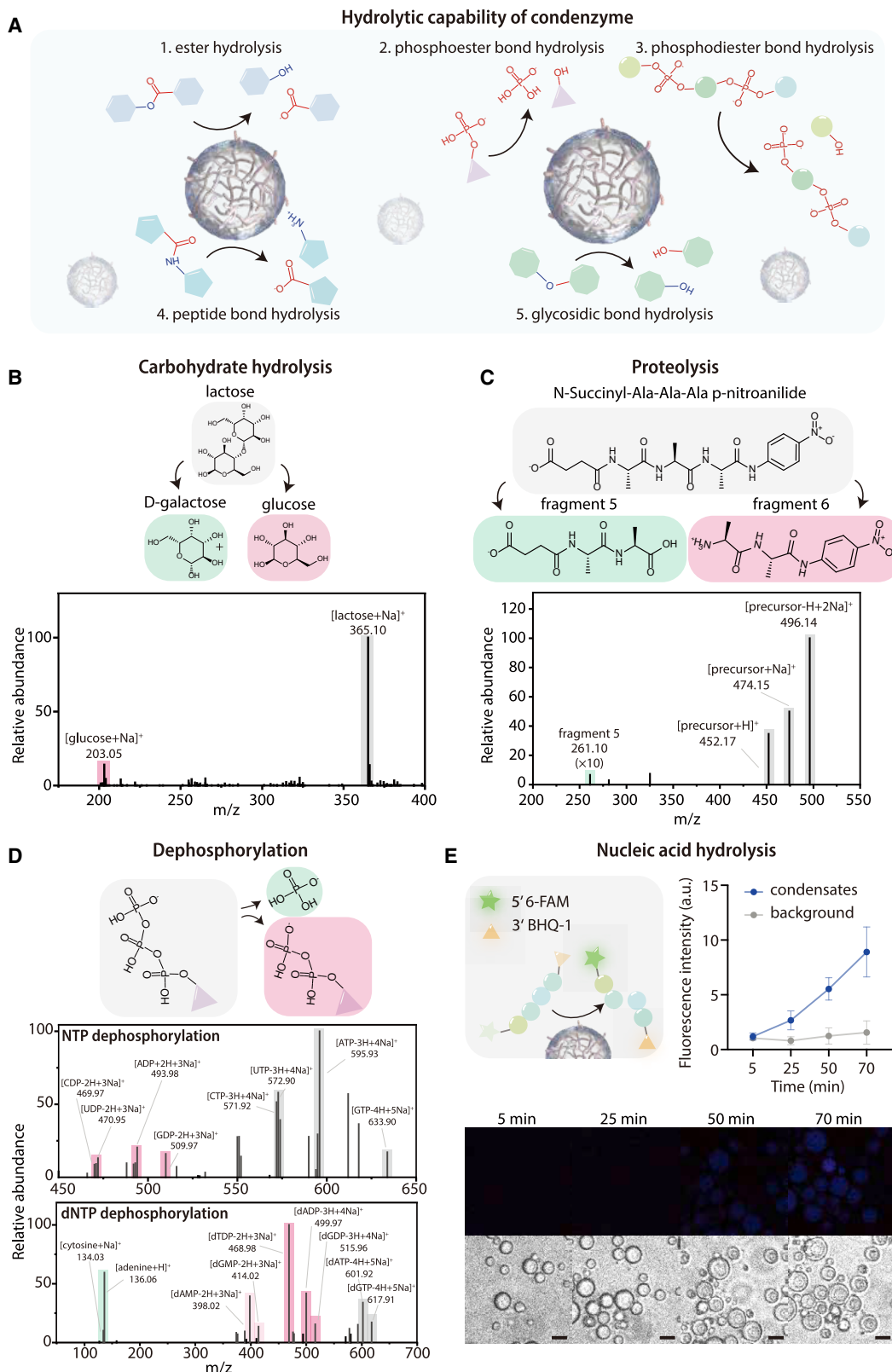
(C) Hyperspectral SRS mapping of water heterogeneity in condensates with ratiometric SRS images ($I(3,650 \text{ cm}^{-1})/I(3,450 \text{ cm}^{-1})$) shows the relative abundance of free water to distorted water. Scale bar, 3 μm .

(D) Hyperspectral SRS mapping of water heterogeneity in condensates with images shows the averaged OH frequency $\langle\omega\rangle$. Scale bar, 3 μm .

(E) Quantitative analysis of averaged OH frequency $\langle\omega\rangle$ for the interfaces of different condensates. Statistical significance was analyzed by Tukey's HSD test.

(F) Quantitative analysis of averaged OH frequency $\langle\omega\rangle$ for the dense phase of different condensates. Statistical significance was analyzed by Tukey's HSD test.

(G) Quantitative analysis of averaged OH frequency $\langle\omega\rangle$ for the dilute phase of different condensates. Statistical significance was analyzed by Tukey's HSD test.



(legend on next page)

distinguished through their frequency regions (Figure 3A).^{89–91} In general, the red shift of the OH frequency corresponds to a stronger hydrogen-bonding network with more hydrogen bonds between water molecules, while the blue shift means more defects in the hydrogen-bonding network with fewer hydrogen bonds between the water molecules. By quantifying the intensity ratio obtained from the spectrum of different types of hydrogen bonding for all three sequences (RLP_{SKGP}, RLP_{SSGP}, and RLP_{SYGP}), we observed spatial heterogeneity of the local water environment in the condensate systems, where higher populations of free water and lower populations of tetrahedral water were observed at condensate interfaces (Figures 3B, 3C, and S3). We further quantified the overall strength of the hydrogen-bonding network by calculating the average O–H frequency, where the intensity is normalized to the unit area between $\approx 3,100\text{ cm}^{-1}$ and $\approx 3,800\text{ cm}^{-1}$. The maps of the averaged frequency show similar patterns of spatial heterogeneity of the local water environment, where the condensate interface shows a distinct blue shift, indicating a weakened hydrogen-bonding network at condensate interfaces (Figure 3D). Notably, this is reflected as a pattern of anisotropy, a phenomenon specific to interfacial properties,⁹² which are likely due to the alignment of water dipoles at condensate interfaces that are modulated by surface electrostatics and hydrogen-bonding interactions.

To explore whether the sequence-dependent catalytic behaviors are affected by the water activity, we compared the blue-shifted values at condensate interfaces of three distinct condensates by comparing condensates of similar sizes. We observed that the averaged frequency at the RLP_{SKGP} condensate interface is significantly higher than those of the RLP_{SSGP} and RLP_{SYGP} condensates and dilute phases (Figures 3E, 3F, and 3G), suggesting a higher abundance of free water molecules. This aligns with previous work suggesting that weakened hydrogen-bonding networks between water molecules or increased free water/bound water ratios facilitate the participation of water molecules in the catalytic reactions.^{80,81,93} Compared with RLP_{SSGP} and RLP_{SYGP}, the significantly blue-shifted values of the RLP_{SKGP} condensate both inside the condensate and at the condensate interface suggest that the charged amino acid induced considerable disruption to the hydrogen-bonding network of the water microenvironment. The observed chaotropic properties of the lysine residue are consistent with the reported effect from its interaction with water molecules in competition with the water hydrogen-bonding network.⁹⁴ These results collectively confirm that the interfacial electrostatic properties can affect catalytic performance, which involves a synergistic contribution from local water activity modulated by charge-dependent hydrogen-bonding network and interfacial electric fields. The distinct water environments, which can affect the solvation free energy of reactants and products,^{95–97}

further suggest that condensates may reshape the free-energy landscape of chemical reactions, rather than merely modulating activation barriers through electric fields.⁹⁸ This observation implies that conventional thermodynamic frameworks for chemical reactivity may not be directly applicable to reactions governed by changes in water activity and interfacial activities.

Hydrolysis reactions for cellularly relevant substrates

We next asked whether condensates can modulate different types of hydrolysis reactions relevant to cellular processes. Specifically, we investigated the hydrolysis of disaccharides, peptides, NTPs, dNTPs, and nucleic acids with RLP_{SKGP} condensates using mass spectroscopy and fluorogenic assays (Figure 4A). These substrates are common chemicals undergoing hydrolysis reactions critical to metabolic cycles.⁹⁹ For disaccharides and peptides, we studied the hydrolysis of lactose and N-Succinyl-Ala-Ala-Ala-*p*-nitroanilide. We discovered that condensates hydrolyzed lactose into glucose and galactose, while we did not find any product in the subsaturated RLP_{SKGP} solution ($C_{\text{RLP}} < C_{\text{sat}}$) (Figures 4B and S4A). For succinyl-Ala-Ala-Ala-*p*-nitroanilide hydrolysis, compared with the subsaturated RLP_{SKGP} solution, we found that condensates enhance the hydrolysis rate over 15.8-fold (Figures 4C and S4B). For the hydrolysis of NTP and dNTP, we found that condensates are capable of hydrolyzing all types of NTPs, generating dephosphorylated products with collective relative abundances exceeding 50% of their precursors, whereas NTPs in solution without condensates were largely preserved in the same reaction conditions. Similar results were observed for dNTPs (Figures 4D and S4C). Lastly, we studied the hydrolysis of nucleic acids by employing a single-stranded DNA (ssDNA) containing a dye-quencher pair, 6-FAM and BHQ-1, at its 5'- and 3'- ends (Figure 4E), the hydrolysis of which breaks the phosphodiester bond and leads to the release of fluorescent 6-FAM. Compared with subsaturated solutions without condensates or in buffer alone, we observed time-dependent increases of fluorescence in solutions containing condensates (Figures 4E and S4D). This observation was further verified with mass spectroscopy analysis (Figure S4E). The capability to hydrolyze nucleic acid by condensates was further confirmed using a supercoiled plasmid as a substrate (Figure S4F).¹⁰⁰ These observations suggest that the hydrolytic activities of condensates are generalizable to different biologically relevant substrates and that these are strictly condensate-dependent reactions.

Generality of hydrolysis by different condensates with substrate-dependent reactivity

We next asked if condensates formed by different types of biomolecular compositions also possess catalytic functions. To explore whether the catalysis of hydrolysis by condensates

Figure 4. Characterization of the generality of hydrolytic capability of condensates toward disaccharide, peptide, NTPs, and nucleic acid

- (A) Schematic demonstration of the hydrolysis reactions catalyzed by the RLP_{SKGP} condensates, including esterolysis, dephosphorylation, nucleic acid hydrolysis, peptide hydrolysis, and disaccharide hydrolysis.
(B) Mass spectroscopy analysis of the hydrolysis of glucose.
(C) Mass spectroscopy analysis of the hydrolysis of peptides.
(D) Mass spectroscopy analysis of the hydrolysis of NTP and dNTP.
(E) Mass spectroscopy and fluorogenic assay of the hydrolysis of ssDNA. Scale bar, 5 μm .

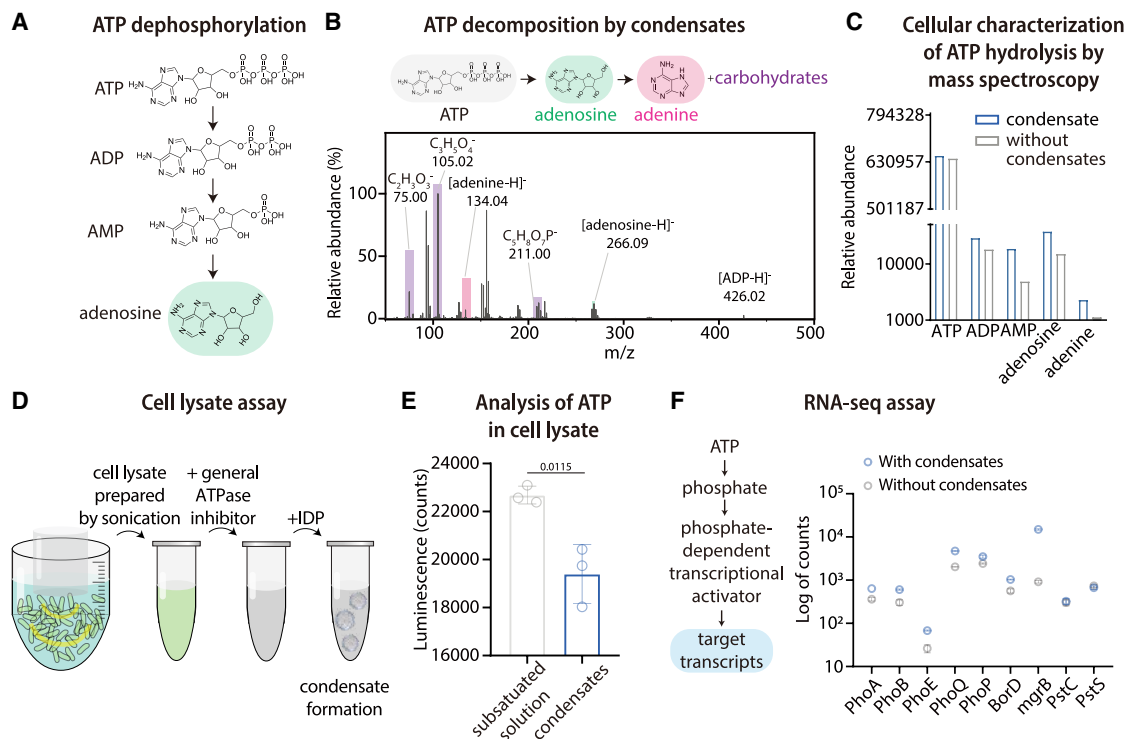


Figure 5. Characterization of the hydrolytic capabilities of the condensome in cells and their biological implications

(A) Schematic illustration of the dephosphorylation of ATP into ADP, AMP, adenosine, and adenine.

(B) Mass spectroscopy analysis of *in vitro* hydrolysis of ATP by the RLP condensates.

(C) Mass spectroscopy analysis of the cellular ATP, ADP, AMP, adenosine, and adenine levels in cells with (60 min after IPTG induction) and without condensates (40 min after IPTG induction).

(D) Schematic illustration showing the protocol for assessing the hydrolytic capability of condensates in cell lysate. Sodium orthovanadate (100 μ M) was introduced to the cell lysate as a general ATPase inhibitor.

(E) ATP assay analysis of ATP levels in cell lysates with RLP condensates or with subsaturated RLP. Lysates were supplemented with 50 μ M ATP for hydrolysis for 15 mins before being quantified. Statistical significance was analyzed by an unpaired *t* test. *N* = 3 independent experiments.

(F) RNA sequencing (RNA-seq) assay analysis of the difference in ATP dephosphorylation-related gene expression in cells with or without condensates. *N* = 2 independent experiments.

might be a generic feature, especially if the IDPs feature ionizable residues, which are essential for generating interphase ion gradients,¹⁷ we reconstituted different condensates formed by IDRs of the dead-box helicase 4 (Ddx4) from nuage bodies,¹⁰¹ the prion-like low-complexity domain of hnRNP A1 (A1-LCD),⁶⁶ the nucleolar protein Ly-1 antibody reactive (LYAR),¹⁸ and the elastin-like polypeptide (ELP).¹⁰² We used the pNPP assay to evaluate whether condensates formed by these IDPs can catalyze reactions. We found that compared with the buffer solution and the subsaturated protein solutions, the solutions containing condensates formed by Ddx4 and LYAR were able to catalyze the decomposition of pNPP. Conversely, A1-LCD and ELP condensates showed minimal functionality as catalysts toward pNPP hydrolysis (Figure S4G). We also reconstituted complex coacervates formed via heterotypic protein-RNA interactions using nucleophosmin 1 (NPM1) and mature rRNA¹⁸ (mat-rRNA).¹⁸ With the pNPP assay, compared with the solutions containing only NPM1 protein, mat-rRNA, and buffer, only the solution containing the NPM1-rRNA condensates was able to accelerate the decomposition of pNPP (Figure S4G). These ob-

servations suggest that the hydrolytic ability of condensates is encoded by the compositions of their biomolecular constituents, which define the interphase electric potentials of different condensates under equilibrium. Our findings regarding the catalytic activities of NPM1 + mat-rRNA condensates are likely to have broad implications for activities at the interface between granular components of nucleoli and the nucleoplasm.

Additionally, we assessed the hydrolysis activities of different condensates using ATP, which is an important metabolite that contributes to dephosphorylation reactions in cells. We discovered that all these condensates (including those that could not hydrolyze pNPP) mediate ATP hydrolysis, resulting in significant decreases in ATP concentration in the solution (Figure S4H). These observations suggest that the chemical property of the substrate dictates the rate of condensate-mediated hydrolysis.

Condensates can fully decompose ATP

Given the significance of ATP hydrolysis in energy generation and cellular signaling, we next focused on studying the hydrolysis of ATP by condensates (Figure 5A). ATP hydrolysis is a

process typically mediated by ATPases¹⁰³ in which ATP is decomposed to ADP through dephosphorylation reactions.¹⁰⁴ After incubating condensates with 50 μ M ATP for 15 min at room temperature, we found that over 30% of the ATP was decomposed by condensates (Figure S5A). To analyze the products of the decomposition reaction of ATP, we employed high-resolution linear ion mass spectroscopy. Surprisingly, we not only found the existence of ADP and AMP, but we also detected distinct peaks at m/z ratios of 134.05, 105.02, and 75.00 that correspond to adenine and different carbohydrates (Figure 5B). This surprising finding suggests that the RLP_{SKGP} condensates facilitate the dephosphorylation of ATP, and they contribute to the bond breakage of adenosine to form adenine and different carbohydrates. This bond breakage might be attributed to the free radicals generated by condensates through their spontaneous redox activities.^{8,105,106} These results further highlight the fact that condensate-specific microenvironments and interfacial properties jointly contribute to the ability of condensates to catalyze reactions.

Next, to evaluate the ATP hydrolysis activities of condensates in live cells, we transformed *E. coli* BL21 cells (DE3) with an isopropyl β -D-1-thiogalactopyranoside (IPTG) inducible pET24 plasmid encoding the RLP_{SKGP} under a T7 promoter.⁴⁶ We implemented non-targeted mass spectroscopy to map the change in ATP hydrolysis pattern in cells with or without condensates during their exponential growth phases (see STAR Methods for treatment details). Experimentally, cells with or without condensates were both induced by the same concentration of IPTG after 90 min of culture; cells with condensates were induced for 60 min and cells without condensates were induced for 40 min, and both samples were harvested in the exponential growth phase.

We discovered that RLP condensates led to an increase of ATP hydrolytic products in cells, such as the increase of intracellular AMP and adenosine (Figure 5C). This observation aligns with our observations *in vitro* (Figure 5B), suggesting that RLP condensates actively drive ATP decomposition in living cells. To further investigate whether this ATP decomposition is condensate-dependent, we compared the ATP decomposition pattern (product/reactant ratio: (ADP + AMP + adenosine + adenine)/ATP¹⁰⁷) between cells induced to express RLP and cells induced to express green fluorescent protein (GFP)-soluble IDP, a fusion protein with a large disordered protein but unable to phase separate.¹⁰⁸ Notably, a dramatic increase of the (ADP + AMP + adenosine + adenine)/ATP ratio was found in cells with the formation of RLP condensates, while this ratio remained stable in cells expressing GFP-soluble IDP with the same culturing and inducing condition (Figure S5B). These observations collectively suggest that the formation of condensates can promote ATP hydrolysis in cells, and induced expression does not account for the generation of ATP decomposition products.

As an additional control, we leveraged the leaky expression nature of the T7 promoter to track the correlation between condensate formation and the decrease of cellular ATP levels, thereby fully excluding the possible metabolic burden caused by overexpression on the consumption of ATP.^{109,110} We applied phase contrast imaging to track condensate formation and correlated the results with the time-dependency of ATP levels of cells. We observed a gradual increase of ATP levels alongside

exponential cellular growth before condensate formation and a sharp decrease of ATP levels at a time point that aligns with condensate formation (Figure S5C). Furthermore, such a decrease of ATP was not observed in cells with leaky expression of GFP-soluble IDP (Figure S5D). This process was studied within a short period and in a rich medium, thereby eliminating the possibility of nutrient limitations.

To further isolate the hydrolysis contribution of condensates in complex cellular environments, we prepared the *E. coli* cell extract lysates¹¹¹ and treated the lysates with sodium orthovanadate, which is the inhibitor of the *E. coli* ATPase activity,¹¹² thus generating a cytoplasm-like environment with limited endogenous enzymatic hydrolysis activities (Figure 5D). We verified that compared with the subsaturated RLP condition in cell lysate, the existence of RLP condensates substantially enhanced the ATP hydrolysis behavior (Figure 5E). This assay verified that the hydrolysis activity by condensates is still robust in complex cellular environments.

To analyze the generality of the endogenous cellular activities, we tested ELP condensates in *E. coli* using an established protocol.^{19,102} With the same set of assays and control experiments used for RLP condensates in *E. coli*, we also proved that the ELP condensates could drive ATP hydrolysis in living cells (Figures S5E–S5H). These experiments demonstrate that the ATP hydrolysis activity of condensates is valid in living cells.

Condensate-mediated ATP decomposition activates phosphate-dependent cellular pathways

To evaluate whether the condensate-mediated ATP decomposition can activate specific cellular pathways, we implemented RNA sequencing to quantify if and how the targeted transcriptome correlated with the consequences of ATP decomposition. We specifically assessed the free phosphate-dependent cellular pathways in *E. coli*,^{113–117} in which free phosphate binds to the transcription factor and activates downstream gene expression. We prepared cells with or without condensates from log-phase culture and analyzed them using RNA sequencing. We found that the gene clusters driven by phosphorylated PhoB and PhoP transcription factors^{115,118,119} are significantly upregulated (p value < 0.048) when comparing cells with condensates and cells without condensates (Figure 5F). In a separate experiment, we found that ELP condensates were also able to activate certain pathways (Figure S5I). The shift of RNA profiles suggests that the catalytic functions of condensates generate global changes of cellular chemical landscapes that can potentially affect cellular pathways.

Catalytic functions of condensates modulate intercellular signaling

The robust capability of condensates to modulate intracellular catalytic reactions suggests a potential role of condensates on cellular signaling through their inherent catalytic functions. To test for this possibility, we first implemented two orthogonal sets of intracellular fluorogenic hydrolysis assays using fluorescein-di-beta-D-galactopyranoside (FDG) and fluorescein diphosphate (FDP), the hydrolysis of which results in the generation of fluorescein. The capability of RLP condensates to hydrolyze FDG and FDP probes was first verified *in vitro* (Figures S6A–S6D).

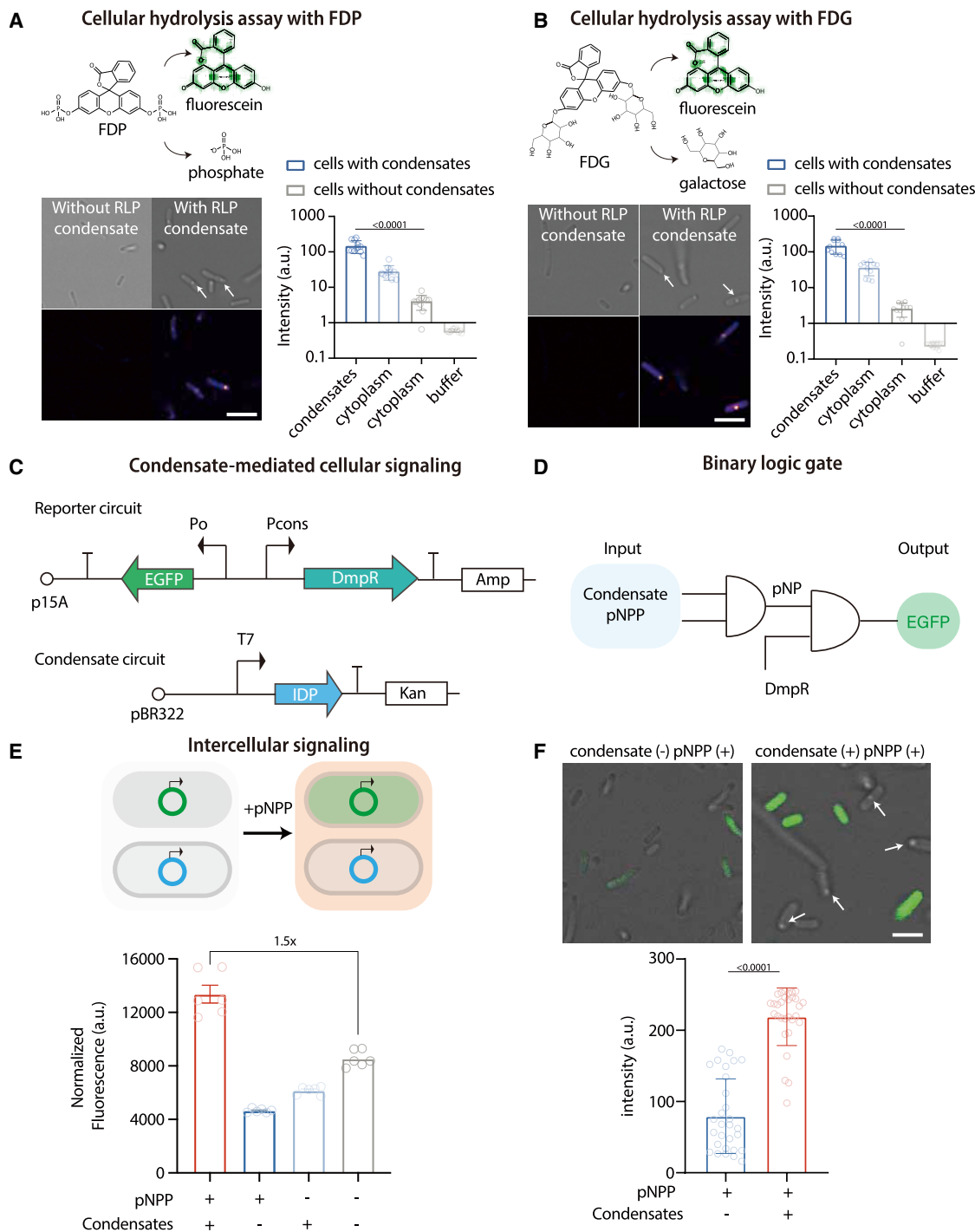


Figure 6. Catalytic functions of condensates mediate the activation of synthetic gene circuits in *E. coli*

(A) FDP probe-based fluorogenic hydrolysis assay analysis of the hydrolysis capability of cells with condensates (60 min after IPTG induction) and without condensate (40 min after IPTG induction). White arrows point to the condensates in the phase contrast imaging. Scale bar, 2 μ m. Statistical significance was analyzed by an unpaired *t* test.

(B) FDG probe-based fluorogenic hydrolysis assay analysis of the hydrolysis capability of cells with condensates (60 min after IPTG induction) and without condensate (40 min after IPTG induction). Scale bar, 2 μ m. Statistical significance was analyzed by an unpaired *t* test.

(C) A reporter circuit was built using DmpR, a phenol-activated transcription activator as a sensor, to drive the expression of EGFP. A condensate circuit was built to generate the cellular signaling event, which is mediated by condensate-mediated pNPP degradation to produce phenol.

(legend continued on next page)

Subsequently, we incubated the probes with cells with or without RLP condensates in their exponential phases at the same cellular density. After 30 min of incubation, compared with cells without condensates, we found dramatic increase of the fluorescein signal in the cells with condensates (Figures 6A and 6B). We also compared the catalytic behaviors of ELP condensates and RLP condensates in cells and found that ELP condensates showed limited catalytic activity (Figures S6E and S6F). This result corroborates our previous conclusion that the hydrolytic capacity of condensates is collectively governed by both condensate components and substrate chemistry.

Additionally, since FDG is hydrolyzed exclusively by LacZ,^{45,120} we next implemented the KrX *E. coli* strain with the LacZΔM15 mutation¹²¹ to test whether the expression of RLP condensates could hydrolyze FDG under a cellular condition without any endogenous enzymatic activity. Indeed, the generation of fluorescence signal correlates with the formation of condensates (Figure S6G). This observation confirmed that condensates can drive hydrolytic activities in living cells independently of endogenous enzymes.

These endogenous hydrolytic activities suggest that condensates can transform signaling molecules, which can serve as a potential strategy to modulate intercellular signaling for synthetic biology.^{122–126} We next designed and engineered a synthetic gene circuit that can be activated by the catalytic product of condensate-mediated hydrolysis reactions. We elected to use the phenol product generated from the pNPP reaction to activate a gene circuit (Figure 6C). To connect the phenol generation with a cellular response, we implemented a σ_{54} -dependent phenol-responsive transcription regulator DmpR,¹²⁷ which specifically drives the activation of the phenol degradation pathway in bacteria,¹²⁸ to regulate the expression of an enhanced GFP (EGFP) (Figure 6D). This serves as the downstream reporter, thereby establishing a logical AND gate signaling cascade initiated by the chemical activity of condensates. We confirmed the activity of the DmpR circuit by monitoring the EGFP response in cell cultures with or without the phenol product. The cell culture with 100 μ M phenol showed a 3-fold higher normalized EGFP signal compared with the culture without phenol (Figure S6H).

We explored whether the catalytic activity of condensates could be used to control different populations of cells based on intercellular signaling. We transformed the *E. coli* BL21 (DE3) cells with the plasmid encoding the RLP_{SKGP} as the signaling cell and the *E. coli* DH5 α with the reporter circuit as the sensor cell. We then co-cultured the two populations of cells in one culture (Figure 6E). Specifically, we incubated log-phase BL21 (DE3) cells with or without condensates with DH5 α cells in M9 medium containing pNPP in an equal volume fraction and tracked the activated EGFP signal. We found that only the

BL21 (DE3) cells with condensates could activate the EGFP signal in the DH5 α cells (Figure 6F). To test the robustness of the signal, we varied the volume fraction between the BL21 (DE3) cells with condensates and the DH5 α cells under a fixed total cell density. We found that increasing the fraction of BL21 (DE3) cells in the multicellular population could further increase the EGFP signal (Figure S6I), even though the absolute quantity of DH5 α cells was decreased. This observation suggests that the rate-limiting step for the intercellular signaling was the activation of gene circuits.

DISCUSSION

In this work, we showed that biomolecular condensates can catalyze hydrolysis reactions. Our data are consistent with condensate-specific electrochemical properties being the defining features of condensates^{8,9,16–18,24,129} that enable them to function as mesoscale enzymes. Accordingly, we refer to condensates that lack biomolecules with intrinsic catalytic activities but show abilities to function as catalysts as being condenszymes. In enzymes, the creation of catalytic environments is made possible by the chemistry that is present in active sites.^{42,54} For condenszymes, the overall electrochemical environments and the interfacial electric fields, which are encoded in the solvent environments defined by the condensed biomolecules through phase transition, drive their catalytic behaviors.

A diverse set of cellularly relevant molecules can be hydrolyzed by condensates, including disaccharides, peptides, NTPs, and nucleic acids. Each of these substrates involves different types of chemical bonds available for nucleophilic attack by water molecules. A surprising finding is that the chemical function of condensates goes beyond hydrolysis. The RLP-based condensates can catalyze the full decomposition of ATP. In sharp contrast to the hydrolysis of ATP by ATPase, which catalyzes the decomposition of ATP to ADP or AMP, the condensates we studied are capable of decomposing ATP all the way to adenine, which requires the removal of the ribose sugar molecule from adenosine. This appears to be realized by free hydroxyl radicals generated by the ability of condensates to drive spontaneous redox reactions,^{8,106} which can lead to the release of ribose.¹³⁰ Given the highly reactive nature of free radicals, these findings suggest that compared with the well-characterized enzymatic pathways, condenszymes might produce unconventional chemical products that may have been long embedded in the unknown metabolites within the metabolic networks.^{131,132} Using a synthetic biology approach, we demonstrated that the catalytic functions of condensates are valid in living cells. Moreover, such chemical functions are robust enough to lead to global effects in a cell and in a population of cells. This highlights

(D) The designed gene circuit follows a binary logic gate cascade with the formation of condensates and the existence of pNPP as the inputs to generate pNP, which activates DMPR to drive the expression of EGFP protein as the output.

(E) Evaluation of the capability of condensates to mediate intercellular signaling with the condensate circuit transformed into the *E. coli* BL21 (DE3) cells and the reporter circuit transformed into the *E. coli* DH5 α cells. The fluorescence signal was normalized to the cellular density. $N = 6$ independent experiments. Bar graph shows the mean $\pm$ SD.

(F) Confocal microscopy analysis of GFP expression in *E. coli* DH5 α cells containing the reporter circuit, co-cultured with pNPP and *E. coli* BL21 (DE3) cells with or without RLP condensates. White arrows point to the condensates indicated by the phase contrast imaging. Scale bar, 2 μ m. Statistical significance was analyzed by an unpaired t test.

the fact that condensates are unlikely to be just passive storage depots that are in any way incidental to cellular functions.¹³³ Instead, we reason that the catalytically active nature of condensates has a direct influence on the regulation of biochemical landscapes in cells.

We deployed a series of IDPs of native proteins, including those from Ddx4, hnRNP A1, and LYAR. We found that the condensates formed by charge-rich IDPs, including the RLPs and IDRs from Ddx4 and LYAR, were robust in catalyzing reactions, while condensates formed by IDPs with lower charge contents, such as the A1-LCD and the non-charged ELP, were less capable of catalyzing reactions. This can be rationalized by the possibility that charge contents of IDPs correlate with the extent of asymmetric ion distribution upon phase transition.^{28,134–141} This conjecture is supported by the active catalytic behavior of the condensates formed by complex coacervation between NPM1 and mat-rRNA.

Asymmetrical partitioning of solution ions across the coexisting phases defines a chemical environment for the dense phase that is distinct from the dilute phase.^{16–18} This gives rise to interphase electric potentials that are essential for generating interfacial electric fields.^{16–18} As native condensates, especially RNP granules are enriched in charged biomacromolecules,^{3,4,39,40,142} such as RNAs and oppositely charged proteins, the asymmetric partitioning of ions and protons will set up distinct microenvironments and interfacial electric fields that provide the impetus for the biochemical activities of condensates.^{8,31,32}

Given the ubiquitous nature of condensates in living cells, we propose that condensation functions might be critical to understanding biogenesis and chemical homeostasis. Biogenesis requires cascading biochemical reactions that also mandate the presence of a chemical sink. In this case, the spontaneous catalytic activity of condensates, which can drive the hydrolysis of diverse substrates, can initiate the cascade by defining a continuous baseline chemical activity. Further, hydrolysis reactions can diversify the chemical landscapes and provide thermal energy to drive other reactions, which are also favorable for the increased chemical complexity during biogenesis. Chemical homeostasis, which suggests the maintenance of stable levels of chemicals in living cells, requires balanced consumption and production of chemicals.¹⁴³ Such processes can be achieved through chemical oscillation,¹⁴⁴ which requires a constant negative feedback to maintain the chemical driving force. The capability to break chemicals through hydrolysis or radical-dependent reactions of condensates can ideally enable negative feedback to maintain a stable chemical synthesis and consumption rate. Such driving forces in the case of transcription and translation condensates might serve as the key factor maintaining stability and control over gene regulatory circuits. These functional aspects of condensates further differentiate them from enzymes or ribozymes, which function at the level of individual molecules and act on specific targets. Because catalytic rates are typically defined as reaction rates per unit of catalyst, direct comparisons between condensates—assemblies composed of many molecules—and single enzymes are not straightforward. The development of techniques for single-condensate analysis could open avenues to establish

condensation profiles,¹⁴⁵ linking composition-dependent electrochemical environments to quantitative catalytic behavior.

What are the factors encoding the catalytic functions of condensates? A key feature driving the catalytic reaction is the interfacial electric field, which is the electric double layer defined as the Gibbs dividing plane in the interphase region where charge separation occurs.^{23,146–148} First, under electrochemical potential equilibrium, the differential partitioning of ions sets up an electric potential gradient between phases thus enabling the charge asymmetry required for setting up the double layer.^{8,17,19,22,24,136,148,149} In the case of biomacromolecule phase separation, the ion gradient can be established through ionizable residues on the biomacromolecules; however, the hydration preferences of the backbones of biomacromolecules can alter the local water environment, which in turn can affect the partitioning of ions based on the free energy of hydration of ions.¹⁵⁰ Therefore, in theory, any kind of asymmetry caused by phase transitions,²³ such as differences in water structure, water abundance, or ion abundance between phases or even limited chain flexibility, should be able to create a solvent environment that can set up an electric potential gradient.^{19,151} Second, the presence of a physical surface in the solution results in anisotropic forces acting on the solvent,^{59,152,153} and this can influence the net orientation of the solvent dipoles and the alignment of excess charges at the surface of condensates. Therefore, the presence of a surface, which drives the change of solvation environment,^{23,30} can also enable the formation of an electrically active interface. Further, the emergence of an electric field at a surface is governed by various mechanisms, suggesting its widespread occurrence.^{59,154} These factors might also support the idea that the catalytic activities should be generalizable for macromolecular assemblies.^{30,155} Further, the microenvironments within the dense phase encode distinct ionic environments, micropolarity, and water activities,^{11,17,156} thereby modulating various factors that can be critical to the chemical activity of condensates. Lastly, beyond microscopic structures, phase transitions can also drive the formation of subsaturated clusters in the dilute phase.^{73,74} This might possess similar properties as micelles or microphases that can also drive spontaneous catalytic activities.¹⁵⁷ Understanding distinct phase transition processes should further enhance our knowledge on the functional landscape of biomolecular structures.

A role of water: Besides the asymmetric distribution of water molecules, we infer, based on our experimental characterizations, that the hydrogen bonding of water is heterogeneous in the condensates. More free water molecules exist at the interfaces of condensates, which supports our hypothesis regarding the roles of interfaces on modulating the catalytic functions. The distinct water activity can be caused by differences in the interfacial electric field or by variations in surface residues, which may have distinct affinities for water molecules or ions, thereby influencing the water structure. These observations emphasize the idea that condensation not only alters molecular distribution but also molecular activity that is in part contributed by an altered solvation free energy at the interface versus at the bulk. Considering the diverse chemical reactions that water activity and structure play a critical role in,^{158,159} understanding the molecular grammars regulating water will shed light on the chemical functions of condensates.

Can condensymes encode substrate specificity? Though the functions are encoded in mesoscale electrochemical environments, condensates possess distinct interior and interfacial structures based on their molecular compositions.^{64,153} These features can potentially drive specific residues to the interfaces of condensates, similar to the defined surface residues on amyloid assemblies.^{155,160} These residues can encode binding selectivity toward specific chemicals, thereby modulating the specificity of catalytic functions.¹⁶¹ We expect that with enhanced understandings on the molecular structures of condensates,¹⁶² condensate-specific chemical functions will be defined. From the condensate environment aspect, the chemical property of the substrate,^{161,163} such as the polarity of the side chains and the charge condition of the substrate, should govern the initial partitioning of the reactants to the condensates so that understanding condensate environment-substrate interaction will also illustrate how condensymes can encode substrate selectivity.

The functions of condensates have remained elusive, and this has opened the door to criticisms and suggestions that condensates might simply be incidental bystanders in cells.¹³³ Our findings appear to change this picture, with the upshot being that cells must have evolved mechanisms to leverage the inherent catalytic functions of condensymes, or cells have adapted mechanisms that blunt condensyme functions. If the former is true, then working out condensyme functions in cells becomes of utmost importance. If the latter is true, then gains-of-function through condensyme functions of pathological condensates become a challenge for cells, thus opening a chemistry-dependent aspect to think about disease. Our findings add to recent discoveries of condensates having distinct and tunable electrochemical and viscoelastic properties.^{8,16–19,23,29,164–169} While the former discovery shows that condensates can be defined by membrane-like potentials that affect electrochemical equilibria in cells,^{17,19,23} the latter has broad implications for condensates being mechano-sensitive and mechano-responsive elements.^{167,170} The current work adds a biochemical dimension to the electrochemical and viscoelastic considerations of condensate functions. Indeed, our work, integrated with contributions made recently,^{8,16–19,23,29,106,145,164–171} opens the door to thinking about condensates being “coupled electro-chemo-viscoelastic” elements.¹⁷²

Limitations of the study

This study employed *in vitro* and bacterial systems to demonstrate the concept that condensates, formed by biomolecules without intrinsic catalytic activity, can be catalytically active. A limitation is the use of synthetic intrinsically disordered proteins as the IDP platform to study this behavior in a sequence-dependent manner. Although a few native condensates were examined, the systems studied do not fully capture the compositional complexity characteristic of naturally occurring condensates.

RESOURCE AVAILABILITY

Lead contact

Please direct requests for resources and reagents to Yifan Dai (dyifan@wustl.edu).

Materials availability

Plasmids generated in this study are available upon request to the lead contact.

Data and code availability

- Raw data as part of this work are publicly available via the Dai lab GitHub and are available at [10.5281/zenodo.20646178](https://doi.org/10.5281/zenodo.20646178).
- All original code that was developed as part of this work is deposited in the Pappu lab GitHub repository and is publicly available at [10.5281/zenodo.20646097](https://doi.org/10.5281/zenodo.20646097).
- Any additional information is available from the [lead contact](#) upon request.

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AUTHOR CONTRIBUTIONS

Conceptualization: Y.D.; methodology: Y.D., R.V.P., W.M., M.W.C., N.Q., R.N.Z., J.L., M.F., and X.G.; investigation: M.W.C., X.G., N.Q., X.S., M.F., Y.X., A.N., L.Y., Y.M., W.Y., M.R.K., and V.L.; funding acquisition: Y.D., R.V.P., W.M., R.N.Z., and J.L.; supervision: Y.D., R.V.P., W.M., R.N.Z., and J.L.; writing: Y.D., R.V.P., M.W.C., and N.Q.; reviewing & editing: all authors.

DECLARATION OF INTERESTS

The authors declare no competing interests.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

- **KEY RESOURCES TABLE**
- **EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS**
 - Bacteria culture
- **METHOD DETAILS**
 - Expression and purification of IDPs
 - LaSSI Simulations
 - Resorufin phosphocholine-based fluorogenic hydrolysis assay
 - Density functional theory calculations
 - Confocal microscopy for characterizing condensates *in vitro*
 - Isolation of the contribution of interface on the catalytic reactions by decreasing the effective surface areas and number of condensates
 - Isolation of the contribution of interface on the catalytic reactions by adding charged surfactants
 - Hyperspectral SRS imaging
 - *In vitro* ATP hydrolysis assay
 - Hydrolysis of pNPP by different condensates
 - *E. Coli*. intracellular ATP assay
 - Mass spectrometry analysis of *in vitro* hydrolysis reactions
 - Mass spectrometry analysis of the hydrolytic activity of condensates in living cells
 - Hydrolysis of DNA plasmid by RLP condensates
 - Hydrolysis of ssDNA by RLP condensate
 - Condensate-dependent ATP hydrolysis in cell lysates

- RNA sequencing analysis
- FDG/FDP fluorogenic assay for analyzing cellular hydrolysis reactions
- Inter-cellular communications with DmpR circuit
- **QUANTIFICATION AND STATISTICAL ANALYSIS**

SUPPLEMENTAL INFORMATION

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Bacterial and virus strains		
BL21(DE3) Competent <i>E. coli</i>	New England Biolabs	Cat#C2527H
NEB® 5-alpha Competent <i>E. coli</i> (High Efficiency)	New England Biolabs	Cat#C2987H
Single step (KrX) component cell	Promega	Cat#L3002
Chemicals, peptides, and recombinant proteins		
RLP _{SKGP} : resilin like polypeptide (20 repeats of the RLP _{WT} monomer with SKGP as N terminal)	This paper	MWC1
RLP _{SSGP} : resilin like polypeptide (20 repeats of the RLP _{WT} monomer with SSGP as N terminal)	This paper	MWC2
RLP _{SYGP} : resilin like polypeptide (20 repeats of the RLP _{WT} monomer with SYGP as N terminal)	This paper	MWC3
ELP _{VPVG-60} : elastin-like polypeptide (60 repeats of the ELP _{V60} monomer)	This paper	MWC4
DDX4-NT protein	This paper	MWC5
A1-LCD protein	This paper	MWC6
LYAR protein	This paper	MWC7
NPM1 protein	This paper	MWC8
pNPA (4-Nitrophenyl acetate)	Sigma-Aldrich	Cat#N8130
pNPB (4-Nitrophenyl butyrate)	Sigma-Aldrich	Cat#N9876
pNPO (4-Nitrophenyl octanoate)	Sigma-Aldrich	Cat#21742
pNPP (p-Nitrophenyl Phosphate)	Sigma-Aldrich	Cat#20-106
pNP-TMA (4-nitrophenyl trimethylacetate)	Sigma-Aldrich	Cat#135046
Resorufin Phosphocholine	AAT Bioquest	Cat#AATB-11640
Fluorescein diphosphate, tetraammonium salt	AAT Bioquest	Cat#217305-49-2
Fluorescein di(β-D-galactopyranoside)	Sigma-Aldrich	Cat#F2756
Adenosine 5'-triphosphate disodium salt hydrate	Sigma-Aldrich	Cat#A7699
N-Succinyl-Ala-Ala-Ala-p-nitroanilide	Sigma-Aldrich	Cat#S4760
Lactose	Sigma-Aldrich	Cat#17814
Di-4-ANEPPS	Invitrogen	Cat#D1199
5-(and-6)-Carboxy SNARF™-1	Invitrogen	Cat#C1270
4-Nitrophenol	Sigma-Aldrich	Cat#241326
Ribonucleotide Solution Set	New England Biolabs	Cat#N0450S
Deoxynucleotide (dNTP) Solution Set	New England Biolabs	Cat#N0446S
Dimethyl sulfoxide	Sigma-Aldrich	CAT#D8418
Critical commercial assays		
Luminescent ATP Detection Assay Kit	abcam	CAT#ab113849
Deposited data		
Raw data	This paper	https://doi.org/10.5281/zenodo.20646178

(Continued on next page)

Continued		
REAGENT or RESOURCE	SOURCE	IDENTIFIER
Oligonucleotides		
ssDNA hydrolysis probe (5'-FAM-TTT-TTT-TTT-ZEN-TTT-TTT-TTTT-BHQ1)	Integrated DNA Technologies	Y. DAI-407460783-XG PROBE 2
Recombinant DNA		
pET24_RLP _{SKGP}	Dzuricky et al. ⁴⁵	MWC1
pET24_RLP _{SSGP}	This paper	MWC2
pET24_RLP _{SYGP}	This paper	MWC3
pET24_ELP _{VPVG-60}	Quiroz et al. ¹⁶⁵	MWC4
pET24_DDX4 IDP	This paper	MWC5
pET24_hnRNP A1 LCD	This paper	MWC6
His-LYAR (Xenopus)	King et al. ¹⁸	MWC7
His-MBP-Tev-NPM1	King et al. ¹⁸	MWC8
pET24_GFP-soluble IDP	Tang et al. ¹⁰⁰	MWC9
Software and algorithms		
Code for simulation on Figure 2	This paper	https://doi.org/10.5281/zenodo.20646096
MatLab	MathWorks	https://www.mathworks.com/products/new_products/release2023b.html

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Bacteria culture

BL21(DE3) Competent *E. coli* (New England Biolabs) and NEB® 5-alpha Competent *E. coli* (High Efficiency) (New England Biolabs) cells were cultured on 2x YT agar plates supplemented with antibiotic, or in 2x YT liquid medium supplemented with antibiotic and glucose; the type of antibiotic and glucose concentrations are specified in the relevant sections. Single step (KrX) component cells were cultured on 2x YT agar plates with kanamycin, or in 2x YT liquid medium supplemented with kanamycin and 0.5% rhamnose. All agar plate cultures were incubated at 37 °C. Liquid cultures were incubated at 37 °C with shaking at approximately 225 r.p.m. under well-aerated conditions.

METHOD DETAILS

Expression and purification of IDPs

Construction of RLP genes

The RLP_{SKGP} gene was a gift from Chilkoti lab from Duke University. The RLP_{SKGP} gene was modified through site-directed mutagenesis to generate mutants (RLP_{SYGP} and RLP_{SSGP}) at the N-terminus using Q5 Site-Directed Mutagenesis kit (New England Biolabs). The cloned genes were transformed into NEB® 5-alpha Competent *E. coli* (High Efficiency) (New England Biolabs) and selected on a 2x YT plate with kanamycin. A single colony was picked and grown in 2x YT liquid medium (MilliporeSigma) containing 45 µg/mL Kanamycin at 37 °C for 18 h overnight (shaking at ~225 r.p.m.). The cloned plasmids were then purified using QIAprep Spin Miniprep Kit (QIAGEN) and the DNA sequences of the purified plasmids were verified with Sanger sequencing service (Genewiz, Azenta Life Sciences).

Expression of synthetic IDPs

BL21 (DE3) competent *E. Coli* (New England Biolabs) was transformed with RLP genes (RLP_{SKGP}, RLP_{SYGP}, RLP_{SSGP}) or ELP genes (ELP_{VPVG-60}) and the transformed cells were selected by plating with Kanamycin. A single colony was picked and grown in 2x YT liquid medium (MilliporeSigma) containing 45 µg/mL Kanamycin at 37 °C for 18 h overnight (shaking at ~225 r.p.m.). The overnight cell culture was inoculated into 1 L 2x YT liquid medium supplied with 45µg/mL Kanamycin. The bulk culture was first incubated at 37 °C (shaking at ~225 r.p.m.) until the OD₆₀₀ of cell culture reaching 0.5. The expression of IDPs was then induced by adding 0.5 mM final concentration of IPTG. For all the ELP proteins, the cell culture was incubated for another 18 h at 25 °C (shaking at ~225 r.p.m.). For all RLP proteins, the cell culture was incubated for another 18 h at 37 °C (shaking at ~225 r.p.m.).

Purification of RLP proteins

All variants of RLP proteins (RLP_{SKGP}, RLP_{SYGP}, RLP_{SSGP}) were purified using the same protocol. Purification of RLP proteins was modified based on an established protocol using the upper critical solution temperature (UCST) feature of RLP.^{45,173} Each liter of the overnight cell culture was pelleted by centrifugation at 2,000 g for 20 min at 4 °C (Eppendorf CR22N). Each pellet was resuspended in 30 mL Resuspension buffer (50 mM Tris, pH 7.5) and lysed by sonication (4 min total time, each cycle with 2 sec on

and 2 sec off, 30 % amplitude) in an ice bucket. The cell lysate was centrifuged at 20,000 g for 25 min at 4 °C to separate the insoluble and the soluble phases. The insoluble pellet from each liter of bacterial culture was dissolved in 10 mL urea buffer (50 mM Tris, 500 mM NaCl, 8 M urea, pH 7.5) supplied with 500 units Benzonase nuclease (MilliporeSigma) at room temperature for at least 2 h with constantly shaking. The sample was then pre-warmed to 40 °C in a water bath for 30 min and centrifuged at 20,000 g for 25 min at 40 °C. The supernatant was collected and dialyzed into resuspension buffer (50 mM Tris, pH 7.5). The dialysis buffer was changed one time over 3 h. Both the soluble and insoluble fractions in the dialysis bag were collected and centrifuged at 20,000 g for 25 min at 4 °C. The insoluble pellet was dissolved in urea buffer (50 mM Tris, 500 mM NaCl, 8 M urea, pH 7.5) at room temperature for at least 1 h with rocking. The sample was then warmed to 40 °C and centrifuged at 20,000 g for 25 min at 40 °C. The supernatant was collected, and its purity was determined on a SDS-PAGE gel (Bio-Rad, Any kD Mini-PROTEAN TGX Precast Protein Gels). This completes a round of UCST based purification. Another two rounds of temperature-dependent purification were performed before isolating the protein with AKTA size exclusion chromatography (Cytiva). The final purity of the purified protein is higher than 95%.

For liquid stock preparation, the concentration of the RLP stock protein was first adjusted to the concentration below its C_{sat} using the Urea buffer and dialyzed into high-salt buffer (50 mM Tris, 500 mM NaCl, pH 7.5) at room temperature. The dialysis buffer was changed three times over 4 h. The dialyzed sample was first verified its capability to phase separate upon dilution. If unable to phase separate upon dilution, the concentration of the dialyzed sample was concentrated by the Pierce™ Protein Concentrators PES, 10K MWCO (ThermoFisher). Once it is verified, the protein stock was aliquoted into PCR tubes and frozen by liquid nitrogen and stored at -80 °C.

Purification of proteins and RNA for the generation of native condensates

Expression and purification of A1-LCD. BL21-CODONplus RIPL *E. Coli.* cells (Agilent Technologies 230280) transfected with Δ hexa_{His}-TEV-A1-LC 1 were grown in LB broth (Sigma: L3522) in Erlenmeyer flasks with $\geq$ 5-fold head volume at 37 °C (shaking at 220 r.p.m.) until OD₆₀₀ ~0.6 was reached. Cultures were then chilled for 15 min in an ice bath before adding 0.35 mM IPTG. Induced cultures were grown for additional 6 hours at 37 °C. Cells were harvested by centrifugation as described previously for synthetic IDPs, flash frozen, and stored as pellets in 50 mL Falcon tubes at -80 °C.

The cell pellet was resuspended to homogeneity in 35 mL supplemented lysis buffer (50 mM MES, 500 mM NaCl, 14.3 mM BME (β -Mercaptoethanol), 200 μ M PMSF (phenylmethylsulfonyl fluoride) pH 6). Lysis buffer was then supplemented with: 500U DNaseI (Sigma: 4536282001), 500U RNase A (Sigma: 10109169001), 5 mg Lysozyme (Sigma: 62971-10G-F), and one protease inhibitor tablet (Sigma: 40694200). The cell suspension was lysed via sonication on a Branson 550 sonicator with an L102C horn attachment using five series of the following 20-round cycle: 1 second on / 2 second off at 30% power. The lysate was then pelleted through centrifugation for 30 min at 38,000 g. The lysate pellet was resuspended in a resuspension buffer (6 M GdmCl, 20 mM Tris, 15 mM imidazole, 14.3 mM BME, and 200 μ M PMSF, pH 7.5) via repeat pipetting and pulses of sonication until the solution was visibly homogenous. This solubilized pellet was centrifuged at 38,000g for 30 minutes at room temperature and the supernatant was applied to a gravity column with 5 mL bed volume of HisPur NiNTA resin (ThermoFisher Scientific) equilibrated with resuspension buffer. The Ni-NTA resin was washed with 75 mL wash Buffer (20 mM Tris, 30mM imidazole, 4 M urea, 14.3mM BME, and 200 μ M PMSF; pH 7.5), then protein was eluted in elution buffer (20 mM Tris, 350mM imidazole, 4 M urea, 14.3mM BME, and 200 μ M PMSF; pH 7.5). Peak fractions from Ni-NTA affinity purification were pooled and diluted 1:1 in dilution buffer (20 mM Tris and 14.3 mM BME; pH 7.5). To this, 1 mg of TEV protease was added, and the mixture was dialyzed overnight in dialysis buffer (20 mM Tris, 2 M urea, 50 mM NaCl, 0.5 mM EDTA, and 1 mM DTT; pH 7.5) at room temperature. After 16 h, the protein solution was filtered through a 0.22 μ m filter, then further purified via ion exchange chromatography on an AKTA Pure fast protein liquid chromatography (FPLC) module using a HiTrap SP 5mL column (Cytiva: 17115201). The column was equilibrated with buffer A (20 mM Tris, 2 M urea, 50 mM NaCl, and 14.3 mM BME; pH 7.5) and protein was bound then subjected to a contiguous gradient protocol with from 0.05 M NaCl to 1 M NaCl. Fractions containing A1-LCD were pooled then further purified and buffer exchanged using size exclusion chromatography - HiLoad 16/600 Superdex 200pg column (Cytiva: 28989335) into the following storage buffer: 20mM MES and 4 M GdmCl at pH 5.5. Lastly, protein was pooled and concentrated in Amicon Ultra 3 KDa MWCO (molecular weight cut-off) concentrator columns (MilliporeSigma: UFC 500396), according to the manufacturer's suggestions. Concentrated protein was stored at 4°C for further usage.

Expression and purification of DDX4-NT. BL21 *E. coli.* cells (NEB) transfected with His-SUMO-DDX4-NT 2, a gift from Professor Lewis E. Kay, were grown in the same manner as A1-LCD. Cells were lysed in the same manner as A1-LCD with a different lysis buffer (20 mM sodium phosphate, 750 mM NaCl, 20 mM imidazole, 14.3 mM BME, and 0.2 mM PMSF; pH 7.5). The supernatant was spun for 25 min at 38,000g and transferred into an equilibrated HisTrap FF Crude 5mL column (Cytiva: 11000458) with an AKTA Pure fast protein liquid chromatography (FPLC) module. The Ni-NTA column was washed in 75 mL lysis buffer then protein was eluted in elution buffer (0.02 M Sodium Phosphate, 0.5 M NaCl, 0.35 M imidazole, 0.0143 M BME, 0.0002 M PMSF; pH 7.5). Peak fractions from this affinity purification were pooled and diluted 5-fold in dilution buffer (0.02 M sodium phosphate, 0.0143 M BME; pH 7.5). This solution was further purified via ion exchange chromatography using a continuous gradient purification protocol with a HiTrap Heparin HP 5mL column (Cytiva: 17040703), Buffer A (0.02 M sodium phosphate, 0.1 M NaCl, 0.0143 M BME; pH 7.5), and Buffer B (0.02 M sodium phosphate, 0.1 M NaCl, 0.0143 M BME; pH 7.5) on the AKTA Pure FPLC module. Peak fractions containing SUMO-DDX4 protein were pooled and cleaved of SUMO tags during an overnight dialysis in the presence of 0.02 ULP1 Protease in cleavage buffer (0.02 M Sodium Phosphate, 0.3 M NaCl, 0.001M DTT (Dithiothreitol), pH 7.5). DDX4-NT was purified to $\geq$ 98% using size exclusion chromatography - HiLoad 16/600 Superdex 200pg column (Cytiva: 28989335) on the AKTA Pure FPLC module in storage buffer

(0.022 M Sodium Phosphate, 1.1 M NaCl, 0.0143M BME, pH 7.5). The DDX4-NT solution was supplemented with 10% glycerol and concentrated in Amicon Ultra 3 KDa MWCO (molecular weight cut-off) concentrator columns (Millipore-Sigma UFC 500396) and concentrated protein was aliquoted into single-use volumes (typically 10 μ L), flash froze in liquid nitrogen, and stored at -80°C . Each step of the purification was assessed in the same manner as described for A1-LCD.

Expression and purification of LYAR. Expression was carried out in the same way as described for A1-LCD. Cells were lysed in supplemented lysis buffer (20 mM sodium phosphate, 500 mM NaCl, 20 mM Imidazole, 14.3 mM BME, 200 μ M PMSF, pH 7). Supernatant was recovered from a 25-minute spin at 38,000g and transferred into a 10 mL volume of equilibrated HisPur Ni-NTA resin (Fisher: 88222) in a gravity column. The Ni-NTA column was washed in lysis buffer until no contaminant protein was detected by Bradford assay. Protein eluted in 20 mL elution buffer (20 mM sodium phosphate, 500 mM NaCl, 350 mM Imidazole, 14.3 mM BME, 200 μ M PMSF; pH 7). Peak fractions from this affinity purification were pooled and diluted five-fold in dilution buffer (20 mM sodium phosphate, 14.3 mM BME; pH 7). This solution was further purified via ion exchange chromatography using a contiguous gradient purification protocol with a His-Trap Heparin HP 5mL column (Cytiva: 17040703), Buffer A (20 mM Sodium Phosphate, 100 mM NaCl, 14.3 mM BME; pH 7), and Buffer B (20 mM sodium phosphate, 100 mM NaCl, 14.3 mM BME; pH 7) on the AKTA Pure FPLC module. Peak fractions containing His-LYAR were pooled and purified to $\geq 95\%$ using size exclusion chromatography - HiLoad 16/600 Superdex 75pg column (Cytiva: 28989333) on the AKTA Pure FPLC module in storage buffer (22 mM sodium phosphate, 1.1 M NaCl, 14.3 mM BME, pH 7.5). The LYAR solution was supplemented with 10% (v/v) glycerol and concentrated in Amicon Ultra 3 KDa MWCO (molecular weight cut-off) concentrator columns (MilliporeSigma: UFC 500396). Concentrated protein was aliquoted into single use volumes (typically 5 μ L), flash frozen in liquid nitrogen and stored at -80°C .

Expression and purification of NPM1. Expression was carried out in the same way as described for DDX4-NT and LYAR. Cells were lysed in supplemented lysis buffer (20 mM sodium phosphate, 500 mM NaCl, 20 mM imidazole, 14.3 mM BME, 200 μ M PMSF, pH 7). Supernatant was recovered from a 25-minute centrifuge at 38,000 g and bound to a 10 mL volume of equilibrated HisPur Ni-NTA resin (Fisher: 88222) in a gravity column. The Ni-NTA column was washed in lysis buffer until no contaminant protein was detected by a Bradford assay. Protein eluted in 20 mL elution buffer (20 mM sodium phosphate, 500 mM NaCl, 350 mM imidazole, 14.3 mM BME, 200 μ M PMSF; pH 7). Peak fractions from this affinity purification were pooled and diluted 5-fold in dilution buffer (20 mM sodium phosphate, 14.3 mM BME; pH 7). Peak fractions containing His-MBP-Tev-NPM1 were pooled and cleaved of MBP during an overnight dialysis in the presence of 0.025 TEV protease in cleavage buffer (20 mM Sodium phosphate, 300 mM NaCl, 1 mM DTT, pH 7.5). This solution was further purified via ion exchange chromatography using a continuous gradient purification protocol with a HiTrap SP 5 mL column (Cytiva: 17115201), Buffer A (20 mM sodium phosphate, 100 mM NaCl, 14.3 mM BME; pH 7), and Buffer B (20mM sodium phosphate, 100 mM NaCl, 14.3 mM BME; pH 7) on the ÄKTA Pure FPLC module. Peak fractions containing NPM1 were pooled and purified to $\geq 95\%$ using size exclusion chromatography - HiLoad 16/600 Superdex 75pg column (Cytiva : 28989333) on the ÄKTA Pure FPLC module in storage buffer (22 mM sodium phosphate, 1.1 M NaCl, 14.3 mM BME, pH 7.5). The NPM1 solution was supplemented with 10% (v/v) glycerol and concentrated in Amicon concentrators. Concentrated protein was aliquoted into single use volumes (typically 5 μ L), flash frozen in liquid nitrogen and stored at -80°C .

Purification of Mature rRNA (mat-rRNA). Mature rRNA (mat-rRNA) is a mixture of rRNAs (18S and 28S) and was purified using the same protocol described previously.¹⁸ Briefly, rRNA was first purified from 50 μ L packed volume of stage VI *Xenopus* oocytes *Xenopus* oocytes via TRIzol reagent extraction (Fisher: 15596026). A pellet of precipitated rRNA was reconstituted in 100 μ L 4°C ddH₂O. Gel electrophoresis was carried out to confirm that the dominate RNA species were the 18s and 28s rRNA. rRNA Concentration was measured by absorbance at 260 nm on a Nanodrop 2000 and immediately aliquoted into single use volumes (typically 5 μ L), flash-frozen in liquid nitrogen and stored at -80°C .

Formation of native condensates

Condensate formation for LYAR and DDX4. To prepare LYAR and DDX4 condensates, we mixed 3 μ L of 2320 μ M DDX4 (in 8 mM NaH₂PO₄, 12 mM Na₂HPO₄, 1 M NaCl) with 27 μ L low-salt solution (8 mM NaH₂PO₄, 12 mM Na₂HPO₄, 0 M NaCl) into a 1.5 mL microcentrifuge tube and incubated for 30 min at room temperature before subjecting to catalytic assay. The corresponding buffer was used as base line to assess the spontaneous hydrolysis activity of the substrate within this buffer.

Condensate formation for A1-LCD. To prepare A1-LCD condensates,⁶⁶ we mixed 27 μ L of 100 μ M A1-LCD (in 20 mM HEPES buffer, pH 7.0) with 3 μ L of high-salt buffer (20 mM HEPES buffer with 3 M NaCl, pH 7.0) into a 1.5 mL microcentrifuge tube with a final concentration of 90 μ M A1-LCD in 20 mM HEPES buffer with 300 mM NaCl. The condensate solution was incubated for 30 min at room temperature before subjecting to catalytic assay. The corresponding buffer was used as base line to assess the spontaneous hydrolysis activity of the substrate in this buffer.

Condensate formation through complex coacervation of NPM1 and rRNA. To prepare NPM1+mat-rRNA condensates, we mixed a 0.1 volume of 200 μ M NPM1 (in mM NaH₂PO₄, 12 mM Na₂HPO₄, 1 M NaCl) with a 0.9x volume of low-salt solution (8 mM NaH₂PO₄, 12 mM Na₂HPO₄) containing mat-rRNA at 11.12 ng/ μ L. The condensate solution was incubated for 30 min at room temperature before subjecting to catalytic assay. The corresponding buffer was used as base line to assess the spontaneous hydrolysis activity of the substrate in this buffer.

Purification of ELP protein

Purification of ELP protein was modified based on an established protocol using the lower critical solution temperature (LCST) feature of ELP.¹⁷⁴ Each liter of the ELP cell culture was pelleted by centrifugation (Eppendorf CR22N) at 2,000 g for 20 min at 4°C . Each pellet from 1 liter of cell culture was resuspended into 35 mL PBS and lysed by sonication (3 min total time, each cycle with 10 sec on and

40 sec off, 75 % amplitude) in an ice bucket. 1 tablet of cOmplete™ Protease Inhibitor Cocktail (Roche) and 500 units of Benzonase Nuclease (MilliporeSigma) were then added into the cell lysate and incubated at room temperature for 1 h. The cell lysate was centrifuged at 20,000 g for 25 min at 4 °C to separate the soluble and the insoluble phases. The supernatant, which contains the soluble ELP, was collected in a new tube. The phase transition of ELP was then induced by adjusting the NaCl concentration to 1 M and heating in a 40 °C water bath for 20 min. The turbid solution, which was phase separated, was then pelleted by centrifugation at 20,000 g for 25 min at 40 °C. The insoluble ELP pellet was physically disrupted using pipette tip and dissolved in 20 mL fresh PBS at 4 °C for at least 3 h with rocking. The dissolved ELP was centrifuged at 20,000 g for 25 min at 4 °C and the supernatant was collected. This completed a full round of ELP purification based on its LCST phase behavior. Another two rounds of temperature-dependent purification were performed before isolating the protein with AKTA size exclusion chromatography (Cytiva). Protein purity was then determined on an SDS-PAGE gel (Bio-Rad, Any kD Mini-PROTEAN TGX Precast Protein Gels). This process would lead to a purity over 95%. The purified ELP was dialyzed into Milli-Q water overnight at 4 °C. The dialysis water was changed three times. The dialyzed sample was then frozen at -80 °C for 1 h and lyophilized for three days to remove all water. The lyophilized dry powder was stored at -80 °C.

Colorimetric assay for the characterization of hydrolysis activity

To prepare RLP proteins for the catalytic activity assay, a frozen protein stock was freshly thawed to room temperature before the experiment. To form RLP condensates ($c > c_{sat}$), thawed RLP stock solution was mixed with a Dilution buffer (50 mM Tris, pH 7.5) with a volume ratio of 3:7. The well-mixed protein solutions were incubated at room temperature for 30 min to allow condensates formation. To prepare subsaturated RLP solutions (no condensates formation at $c < c_{sat}$), RLP stock was diluted by a high-salt buffer (50 mM Tris, 500 mM NaCl, pH 7.5) with a volume ratio of 1:5. The resulting solution was subsequently diluted with a Dilution buffer (50 mM Tris, pH 7.5) at a volume ratio of 3:7.

To prepare ELP condensates for the catalytic activity assay, lyophilized ELP powder was weighted to prepare a stock solution in Dilution buffer (50 mM Tris, pH 7.5). After the powder was fully dissolved, NaCl concentration was adjusted using a high concentration NaCl stock solution (50 mM Tris, pH 7.5, 2 M NaCl) to realize a final salt concentration at 300 mM NaCl.

To prepare chemical stock solutions, a 100× chemical stock (50 mM) was freshly prepared before every experiment. 4-nitrophenyl phosphate (pNPP) was prepared in water. 4-nitrophenol (4-NP), 4-nitrophenyl acetate (pNPA), 4-nitrophenyl trimethylacetate (pNP-TMA) were prepared in acetonitrile. 4-nitrophenol butyrate (pNPB) and 4-nitrophenyl octanoate (pNPO) were prepared in dimethyl sulfoxide (DMSO).

To initiate the catalytic reaction after protein incubation, 2 μL of a 100 chemical stock solution was added per 200 μL of diluted protein sample on a flat-bottom transparent 96-well plate (Corning, COSTAR). For control reaction without chemical, the same volume of corresponding solvent was added. The plate was sealed with a transparent film and immediately transferred to a plate reader for recording. The absorbance was measured at 410 nm and 385 nm with 25 flashes on a TECAN Infinite M200 plate reader or a TECAN Multimode plate reader (TECAN). The reaction was evaluated at 25 °C with a 3-min kinetic interval for at least 70 cycles.

From the measured absorbance at 410 nm for all catalytic reactions, the concentration of 4-nitrophenol, the product of the hydrolysis reactions, was calculated based on a standard curve which was obtained experimentally under the same condition. The baseline of 4-nitrophenol from chemical self-hydrolysis was subtracted from the catalytic reaction. The initial rate (V_0) was calculated from the linear portion of the activity plot ([4-nitrophenol] versus time) by performing a linear regression using Graphpad. For Michaelis-Menten fitting, the means of V_0 values were plotted as a function of substrate concentrations, and the plot was fitted to the Michaelis-Menten equation by using the built-in non-linear regression on Graphpad (Prism). The fitted Michaelis-Menten equation was. All measurements were replicated at least 3 times.

Absorbance scan. The absorbance scan for monitoring pNPA (0.5 mM) hydrolysis reaction catalyzed by RLP_{SKGP} condensates was conducted at room temperature at pH 7.5. The reaction condition and the buffer composition were identical to the aforementioned colorimetric assay. At different incubation time points, measurements were taken using the UV-VIS function on Nanodrop one (Thermo Scientific).

LaSSI Simulations

To evaluate whether the Simulations were performed using LaSSI, a lattice-based Monte Carlo engine.⁶³ Monte Carlo moves are accepted or rejected based on the Metropolis-Hastings criterion so that the probability of accepting a move is equal to $\min(1, \exp(-\beta\Delta E))$, where $\beta = 1/kT$. Here, kT is the simulation temperature, and ΔE is the change in total system energy associated with the attempted move. Total system energies were calculated using a nearest neighbor model and previously derived interaction parameters.²⁵ For each variant and simulation temperature, 300 distinct chain molecules composed of 166 beads each were placed in a cubic lattice with a length of 140 lattice units. Previous calibrations have shown that the numbers of molecules used in these simulations are adequate to avoid problems due to finite size effects.²⁵ The simulations were allowed to equilibrate such that the chains formed a single condensate with a coexisting dilute phase before any analysis was performed. This process was facilitated by initializing the system in a smaller box with side length 40 lattice units. Simulations were performed for 3×10^{10} Monte Carlo steps and analyzed every 5×10^7 steps. Radial densities were determined by calculating the center-of-mass of the system and partitioning the system into radial bins, centered at the center-of-mass, each with radius 1/4 of a lattice unit. Radial densities were normalized by the exact number of lattice sites within each radial bin. Radial densities were further scaled based on the number of beads being analyzed. For example, when determining radial densities for the full-length proteins, densities were divided by 166 (the chain length).

In contrast, when determining radial densities for the N-termini, the first four residues of the protein were used, so the densities were divided by 4. For each set of conditions, three independent simulations were performed.

Resorufin phosphocholine-based fluorogenic hydrolysis assay

The fluorogenic hydrolysis assay was employed to probe (1) the hydrolytic capability of different RLP condensates (RLP_{SKGP}, RLP_{SYGP}, RLP_{SSGP}), and (2) the hydrolytic capability of RLP_{SKGP} condensates upon the change of bulk phase pH. In these experiments, the condensates were prepared as described in the previous section, except (1) RLP variants were used and (2) the Dilution buffers were adjusted to different pH 6, 7, 8 in a specific experiment. After the formation of the condensates, resorufin phosphocholine was added into solutions containing RLP condensates to a final concentration of 50 μ M. The reaction solution was transferred into a 384 well plate (PerkinElmer) and incubated in H301-K stage (okolab) at 30°C. 10% excitation was set at 550 nm with a WLL laser, and the emission detector was set at 570–600 nm with a HYD detector. An autofocus function was set to image a z range of 15 μ m and the imaging layer with the highest fluorescence signal was used for quantification. ImageJ (Fiji) was used for quantification based on Particle Analysis function. Additionally, to evaluate the reactivity of subsaturated RLP solutions as a control, the same concentration of resorufin phosphocholine was also added into subsaturated RLP solutions, and the resorufin fluorescence signal was monitored as described above. To monitor the pH-dependent hydrolysis of resorufin phosphocholine, RLP protein stock buffer (50 mM Tris, 500 mM NaCl, pH 7.5) was first mixed with Dilution buffer adjusted to different pH conditions (pH 6, 7, and 8). Resorufin phosphocholine (50 μ M) was then added to the mixtures. The reaction was monitored as described above.

To verify whether the partitioning of the resorufin affects quantitative comparisons between the fluorogenic hydrolysis activity using different RLP condensates, 50 μ M resorufin sodium salt (Sigma-Aldrich) was added to (1) solutions containing different RLP condensates and (2) solutions containing RLP_{SKGP} condensates prepared with Dilution buffer at different pH conditions. The partitioning of resorufin was monitored via confocal microscopy for 120 min. Resorufin fluorescence signals at different time points in both condensates and dilute phase were quantified using ImageJ (NIH).

Density functional theory calculations

To understand the role of electric field on catalyzing the hydrolysis reactions, we investigated the influence of the external electric field strength on the rate of pNPA hydrolysis computationally using density functional theory (DFT). It has been shown computationally that pNPA hydrolysis is likely concerted with the tetrahedral intermediate being the transition state, and the reaction is well-studied experimentally.^{175,176} In the presence of an electric field, the free energies is modulated by $-\mu \cdot \mathbf{E}$, where μ is the dipole moment of any species and $\mathbf{E}$ is the electric field. The barrier height changes for this reaction due to the electric field is given by

$$\Delta G^\ddagger(\mathbf{E}) = G_{TS} - G_{reactant} = G_{TS}(\mathbf{E} = 0) - G_{reactant}(\mathbf{E} = 0) - \Delta\mu \cdot \mathbf{E} \quad (\text{Equation 1})$$

where $\Delta\mu = \mu_{TS} - \mu_{reactant}$ is the difference in dipole between the reactant and the transition state. From this, the field-induced reaction barrier change is expressed as $\Delta\Delta G^\ddagger = -\Delta\mu \cdot \mathbf{E}$. By employing the Arrhenius equation, the expected rate change was calculated through the equation:

$$\frac{k(\mathbf{E})}{k(\mathbf{E} = 0)} = \exp(\beta\Delta\Delta G^\ddagger) = \exp(-\beta\Delta\mu \cdot \mathbf{E}). \quad (\text{Equation 2})$$

All computations were performed with Q-Chem 6.0.⁵⁵ The dense phase environment was modeled using the conductor-like polarizable continuum model (CPCM), a standard implicit solvation model to model these reaction systems^{177,178}

We investigated a model with an explicit solvent cage¹⁷⁵ along with implicit solvation treatment. We found that without adding an explicit solvent cage, the barrier height was too low (1–2 kcal/mol) for various exchange-correlation functionals and basis sets employed. Therefore, an explicit solvent cage consisting of seven water molecules was added around the reaction complex to shape the transition state geometry.¹⁷⁵ To model the system, the 6–31++G** basis set and B3LYP exchange-correlation functional was first used to compute optimized gas-phase geometries for reactant, transition state, and product. The optimized gas-phase geometries with implicit solvation were studied via CPCM, the def2-TZVPD basis set, and ω B97X-V exchange-correlation function. The solvation-free energies and dipoles for the reaction system were obtained for the reactant, transition state, and product configurations. The dipoles for the reactant and transition state and change in barrier height due to a changing electric field were computed for several dielectric constants ϵ since we cannot measure the precise dielectric constant. Modulating the dielectric constant does not change the qualitative conclusions. We reported the free energy changes based on $\epsilon = 3.2$, the average dielectric constant for proteins¹⁷⁹ and the electric field of 10 MV/cm (a typical value in the protein environment⁵⁶)

Confocal microscopy for characterizing condensates *in vitro*

Analysis of condensate quantity and chemical environments were performed using LEICA STELLARIS 8 FALCON confocal microscopy (LEICA). For characterization of the chemical environments of the condensates, we followed the previous published protocols⁸ using SNARF™-4F 5-(and-6)-Carboxylic Acid (ThermoFisher) and DI-4-ANEPPS (ThermoFisher) dyes for the characterizations of internal apparent pH and the relative strength of interfacial electric fields. Briefly, C-SNARF-4 was added to the samples at a final concentration of 2 μ M. C-SNARF-4 was excited using a 488 nm laser, and emission was collected at 540–600 nm and 610–660 nm. For

ANEPPS, the probe was added at a final concentration of 1 μM . ANEPPS was excited using a 470 nm laser, and emission was collected at 535–545 nm and 610–640 nm. The images were processed through ImageJ with Image Calculator to quantify the ratio of fluorescent signals between each channel. The calibration curve for pH is calculated with the ratio of C-SNARF-4 signal obtained from 150 mM NaCl, 50 mM Tris buffer with different pH.

Isolation of the contribution of interface on the catalytic reactions by decreasing the effective surface areas and number of condensates

To conduct this experiment, the RLP_{SKGP} protein was first purified and diluted to 32 μM in a storage buffer (4 M urea, 500 mM NaCl, 50 mM Tris, pH 7.5). The protein was then dialyzed into a buffer with a lower salt concentration (150 mM NaCl, 50 mM Tris, pH 7.5) to directly trigger the formation of condensates. The dialysis process lasted for 3 hours. The solution containing condensates was then collected for subsequent experiments.

A portion of the condensate-containing solution was centrifuged (20000 g, 20 min) to pellet the condensates, thereby lowering the effective surface area and number of interfaces. Subsequently, the supernatants were collected and filtered through a membrane that allows the passage of molecules with a molecular weight lower than 100 kDa, thereby eliminating nano-size protein assemblies. The solution after this filtering was designated as the dilute phase. Subsequently, half of the dilute phase solution was refilled into the tubes containing the protein condensate pellet. The refilling of the diluted phase was conducted as gentle as possible while ensuring its fully contact with pellets. Finally, pNPP was added to the following solutions: the solution with condensates, the dilute phase solution refilled to the tubes with condensate pellets, the dilute phase itself, and the buffer from dialysis. All solutions were in the same volume, with a final pNPP concentration of 0.5 mM. pNPP hydrolysis in these solutions was analyzed at 30 min, 60 min, and 120 min post reaction. To avoid the interference from the absorbance of the condensates, all samples were centrifuged (13000 g, 3 min, 22 °C) and the supernatants were collected for measurements. The absorbance changes of these solutions at 410 nm were monitored using a plate reader (TECAN).

Isolation of the contribution of interface on the catalytic reactions by adding charged surfactants

To conduct this experiment, RLP_{SKGP} condensate was prepared by diluting the protein stock in high salt buffer (500 mM NaCl, 50 mM Tris, and pH 7.5) with a Tris buffer (50 mM Tris, and pH 7.5) with a volume ratio of 3:7. Upon the formation of condensates, charged surfactants, SDS and CTAB, were supplement into the solution containing condensates with a final concentration of 100 μM , separately. The surfactants were then incubated with condensates for 10 min. Subsequently, ANEPPS assay was conducted to evaluate the effect of charged surfactants on the interfacial electric field of condensates. Further, fluorogenic hydrolysis assay based on the resorufin phosphocholine probe was employed to analysis the effect of charged surfactants on the hydrolytic capability of condensates.

Hyperspectral SRS imaging

Hyperspectral SRS imaging of condensates was acquired by a standard SRS microscope equipped with a picoEmerald FT laser system (Applied Physics & Electronics) to supply a synchronized pump beam and a Stokes beam with a 2 ps pulse width and 40 MHz repetition rate. The Stokes beam is fixed at 1031.2 nm and modulated at 20 MHz. The two beams are spatial-temporally overlapped before being coupled into an inverted laser-scanning microscope (FV1200, Olympus) and focused by x25 water objective (XLPlanN, 1.05 numerical aperture, MP, Olympus) for SRS imaging. The transmitted beams are then collected by the 1.4 NA oil condenser and filtered with a shortpass filter (FESH1000) to leave only the pump beam to be collected by a silicon photodiode (FDS1010, Thorlabs) with a DC voltage of 64 V. The current output from the photodiode was terminated by 50 Ω and demodulated with a high-frequency lock-in amplifier (HF2LI, Zurich Instruments) at 20 MHz frequency to detect the stimulated Raman loss on the pump beam. The in-phase X-output of the lock-in amplifier is fed back into the analog interface box of the microscope to form SRS images.

For mapping the vibrational signature of water OH stretching, the pump beam was tuned in the range between 741 nm and 802 nm. The hyperspectral images were acquired with a pump power of 17 mW and Stokes power of 125 mW on the sample. The time constant at the lock-in amplifier was set at 18 μs , with the pixel dwell time for scanning set to 20 μs .

In vitro ATP hydrolysis assay

To evaluate ATP hydrolysis by condensates *in vitro*, the ATP stock solution (prepared from adenosine 5'-triphosphate disodium salt hydrate) was added to samples to initiate the reaction with a final concentration of 50 μM . The reaction was incubated at room temperature for 15 min before characterization. The hydrolytic capability of native condensates toward ATP was evaluated with the luminescent ATP Detection Assay Kit (Abcam). The ATP quantification was performed following the manufacturer's protocol. In brief, the reaction was terminated by adding the detergent and incubated for 5 min. The treatment of detergents solubilizes macromolecular assemblies,¹⁸⁰ so the partitioning effect of ATP by condensates is excluded (which was verified by mass spectrometry analysis). The substrate reaction solution was then added into the sample and incubated for 15 min in the dark at room temperature. The luminescence was recorded on the Tecan Infinite M200 plate reader with a 2000 ms integration time (Luminescence mode). We evaluated the hydrolytic capability of condensates made of different type of proteins, including DDX4, LYAR, A1 LCD, RLP_{SKGP}, and ELP. The hydrolytic capability of complex coacervated formed by NPM1 and mat-rRNA was also evaluated. For condensate-forming proteins,

the hydrolytic capabilities of their corresponding subsaturated solutions, their condensates, and their corresponding buffers were assayed. For the coacervate group, the hydrolytic capability NPM1, mat-rRNA, and the corresponding buffer were assayed.

Hydrolysis of pNPP by different condensates

The hydrolytic capability of different types of condensates towards pNPP (4 mM) was also analyzed to compare the chemical selectivity. The same condensate and control groups as described above were used for pNPP hydrolysis assay. The pNPP hydrolysis protocol was implemented as described in the colorimetric assay section.

E. Coli. intracellular ATP assay

We quantified the intracellular ATP level in BL21 (DE3) competent *E. coli* based on leaky expression of RLP_{SKGP}, GFP-soluble IDP, or ELP. Experimentally, a single colony of the plated BL21 *E. coli* was picked and grown in 2 mL 2x YT liquid medium supplied with 4 % glucose and 45 µg/mL kanamycin at 37 °C overnight. Each liquid culture was adjusted to OD₆₀₀=0.45 before back dilution at a ratio of 1:100 with fresh 2x YT liquid medium supplied with 45 µg/mL Kanamycin and 4 % glucose, and this is regarded as the time 0 of the culturing. *E. coli* was collected at 120 min, 240 min, 420 min, and 510 min post culturing for subjecting to ATP assay and confocal imaging. ATP levels were quantified using the Luminescent ATP Detection Assay Kit (Abcam) according to the manufacturer's instructions. At each time point, cells were harvested from three independent culture tubes (discarded after collection) and rapidly washed with PBS. The cellular density of the cell suspensions was then adjusted to an OD₆₀₀ at 0.18 in PBS. For each replicate, 300 µL of this suspension was pelleted and resuspended in 30 µL PBS. To lyse the cells and immediately halt ATP consumption, 15 µL of detergent solution was added and incubated for 5 min. Subsequently, 15 µL of substrate reaction solution was added, and the mixture was transferred to a TECAN Multimode plate reader. Samples were subjected to 3 min of shaking, followed by 12 min of dark adaptation, and luminescence was then recorded in Luminescence mode with a 2000 ms integration time.

Mass spectrometry analysis of *in vitro* hydrolysis reactions

The nano-electrospray ionization mass spectrometry (nESI-MS) was employed for the target hydrolytic products detection. The nESI emitter was made of borosilicate glass capillary (Lot: 213425, ITEM# BT-150-10, ID: 0.86 mm, OD: 1.5 mm, Length: 10 cm) and fabricated by a micropipette puller (Sutter Instrument Co., Novato, CA, USA). Major parameters of the pull program were set as follows: HEAT 715; VEL: 15; PULL: blank; TIME: 250; PRESSURE: 720. Sample solution was aspirated and loaded by a micro-loader (Lot# K2080231, 20 µL, Eppendorf, Germany) into the fabricated capillary emitter. A stainless stain needle was inserted from the back side of the emitter to touch the loaded sample solution. The sample-loaded nESI emitter was then positioned in front of a mass spectrometer (Orbitrap Fusion, Thermo Scientific, San Jose, CA, US) inlet at a distance of 3.0 mm. When a +1.5 kV high voltage is applied to the needle, the strong electric field form on the emitter tip will trigger the spraying process for the product compound ionization and MS detection. The target metabolite and its hydrolytic products were monitored by the full scan under a positive mode (*m/z* 100–1000). The MS capillary temperature was set at 275 °C; RF-lens (%) was set at 40, and maximum injection time set at 400 ms. The mass spectrometer was calibrated before each measurement batch by using ion mass calibration solution kits (Pierce positive, Thermo Fisher, San Jose, CA, USA). Before being detected by mass spectrometry, samples were incubated for 4 h at room temperature.

Mass spectrometry analysis of the hydrolytic activity of condensates in living cells

To conduct the mass spectrometry analysis for cellular samples, BL21(DE3) was first transformed with plasmids containing RLP_{SKGP}, GFP-soluble IDP, or ELP. These transformed cells were then plated on 2x YT agar containing 45 mg/L kanamycin and 0.4% w/v glucose. Single colonies were inoculated into 3 mL of 2x YT supplemented with 45 mg/L kanamycin and 0.4% w/v glucose in 14 mL round bottom tubes (Fisher Scientific) and grown overnight in a shaking incubator at 37 °C and under constant shaking at 250 r.p.m. Then, cultures were 1:100 back diluted into fresh M9 media (0.2% Casamino acid, 2 mM MgSO₄ and 0.1 mM CaCl₂) until the exponential phase (OD₆₀₀ = 0.2–0.25) was reached. 3 µL of 0.5 M IPTG was added to the medium, and induced cells were continually cultured for 60 min to allow condensate formation, or 40 min to not allow condensate formation. For the sample collection, all samples were adjusted to the OD₆₀₀=0.4 and 4 mL to so that each group possess same the quantity of cells. Finally, all samples with identical cellular density were centrifuged to obtain cell pellets, which were sent for further processing for mass spectrometry analysis.

For mass spectrometry analysis, cells with identical cellular density between samples were first flash-frozen by liquid nitrogen to quench all biochemical processes within cells. The cell samples were frozen under -80 °C until use. Before MS detection, acetonitrile was added to the concentrated cell samples to deactivate intracellular enzymes and make the final solvent composed of 90% acetonitrile-10% water in volume ratio.^{181,182} The diluted cell suspension was then immediately undergo ultrasonication (Branson 3800 ultrasonic cleaner, 40 KHz) for 5 min to fully disrupt cell membranes and macromolecular structures to release and extract the intracellular metabolites into the solution. Finally, samples were subjected to centrifugation at 5000 r.p.m. to spin down the insoluble cell debris and denatured proteins. Only the supernatant was uploaded into a capillary emitter for mass spectrometry detection.

Hydrolysis of DNA plasmid by RLP condensates

The DNA plasmid was incubated with solutions containing RLP_{SKGP} condensates or buffer for 30 min at room temperature. Subsequently, proteinase K was added to the samples to degrade the RLP_{SKGP} protein for 30 min at 37 °C. The samples were then

subjected to agarose gel electrophoresis to analyze changes in plasmid structure or molecular weight, compared to an untreated plasmid as a reference. Finally, the agarose gel was analyzed using the GelDoc Go imaging system (Bio-Rad).

Hydrolysis of ssDNA by RLP condensate

An ssDNA hydrolysis probe containing 6-FAM at the 5' end and BHQ-1 at the 3' end was purchased from Integrated DNA Technologies. Experimentally, the 6-FAM-ssDNA-BHQ-1 probe (40 nM) was incubated with a solution containing condensates or a sub-saturated RLP solution, or a buffer (150 mM NaCl, 50 mM Tris, pH 7.5). Confocal microscopy was employed to monitor fluorescence emission from the sample at a wavelength of 510–530 nm with an excitation laser at 495 nm. Samples prepared with the same procedure were analyzed by mass spectrometry.

Condensate-dependent ATP hydrolysis in cell lysates

Cell lysates were prepared from wild-type BL21(DE3) cells. Briefly, BL21(DE3) cells were cultured overnight in 2× YT medium. At the second day, cultures were back-diluted (1:100) into 2 bottles (1 L) of M9 medium (0.2% Casamino acid, 2 mM MgSO₄ and 0.1 mM CaCl₂) and grown at 37 °C for 4 h before harvesting by centrifugation at 4 °C. This yielded approximately 4 mL of cell pellet, which was washed twice with a buffer (50 mM Tris, pH 7.5) and resuspended in 4 mL of the same washing buffer. The suspension was frozen at –80 °C for 10 min and then lysed by sonication (2 min total, 2 s on/2 s off cycles, 30% amplitude) in an ice bath. The lysate was clarified by centrifugation (20,000 g, 5 min, 4 °C) three times to remove cell debris, and the resulting supernatant was collected as the separated cell lysate.¹⁸³

Prior to characterize the hydrolytic capability of condensates, sodium orthovanadate (Millipore Sigma) was added to the lysate at a final concentration of 100 μM to inhibit ATPases and phosphatases.^{184,185} The inhibition reaction was incubated for 10 min at room temperature.

For RLP condensates, RLP stock was mixed with cell lysate at a 3:7 volume ratio. Sub-saturated RLP samples were prepared using a 6-fold diluted RLP stock but with the same mixing protocol. For ELP condensates, ELP stock was mixed with lysate at a 2:7 ratio, followed by addition of 3 M NaCl (1:9 volume ratio) to induce phase separation. Sub-saturated ELP samples were prepared using a 6-fold diluted RLP stock but with the same mixing protocol.

ATP hydrolysis was initiated by adding a 100× ATP stock to 30 μL of prepared samples, resulting in a final ATP concentration of 50 μM. Reactions were incubated for 15 min at room temperature before ATP quantification with the luminescent ATP Detection Assay Kit (Abcam) according to the manufacturer's protocol. The quantification procedure is the same as described in a previous section.

RNA sequencing analysis

To conduct the RNA-seq assay, BL21(DE3) was first transformed with plasmids containing RLP_{SKGP} or ELP. These transformed cells were then plated on 2x YT agar with 45 mg/L kanamycin and 0.4% w/v glucose. Upon culturing, single colonies were inoculated into 3 mL of 2x YT supplemented with 45 mg/L kanamycin and 0.4% w/v glucose in 14 mL round bottom tubes (Fisher Scientific) and grown overnight in a shaking incubator at 37 °C and under constant shaking at 250 r.p.m. Then, cultures were back diluted at a 1:100 ratio into fresh M9 media (0.2% Casamino acid, 2 mM MgSO₄ and 0.1 mM CaCl₂) until the exponential phase (OD₆₀₀ = 0.2–0.25) was reached. 3 μL of 0.5 M IPTG was added to the medium. After induction, cultures were incubated for 60 min (condensate-forming) or 40 min (sub-saturated, IDP-expressing but condensate-negative). For the sample collection, all samples were adjusted to OD₆₀₀=0.4 and 4 mL so that each group possesses the same quantity of cells. Finally, all samples were centrifuged to obtain cell pellets, which were sent for RNA-seq analysis at Azenta Life Sciences. The same data analysis procedures were performed as described below by Azenta Life Sciences.¹⁹

FDG/FDP fluorogenic assay for analyzing cellular hydrolysis reactions

Two fluorogenic hydrolysis probes were employed in this experiment: FDP (fluorescein diphosphate) and FDG (fluorescein di(β-D-galactopyranoside)). Both FDP and FDG are well-established probes for analyzing cellular metabolism. The ability of condensates to hydrolyze both probes was first verified *in vitro*. In these experiments, FDP or FDG was added to a solution containing condensates to a final concentration of 1 mM. If either probe is hydrolyzed to generate fluorescein, the fluorescence intensity is expected to increase significantly. This change in fluorescence was monitored using confocal microscopy with an excitation wavelength of 490 nm and an emission range of 510–530 nm. Upon incubation with condensates, the fluorescence intensities in the samples increased, indicating that both probes were hydrolyzed by the condensates.

For the cellular hydrolysis assay, the hydrolysis of FDP and FDG probes was studied using the following cellular conditions: (1) *E. coli* BL21(DE3) cells containing RLP_{SKGP} condensates, BL21(DE3) cells containing RLP_{SKGP} but without condensate formation, and BL21(DE3) cells with ELP condensates; and (2) *E. coli* K12 cells LacZΔM15 containing RLP_{SKGP} condensates or the RLP_{SKGP} without condensate formation. The overnight cell cultures were prepared based on the same protocol as described above. All groups were induced by IPTG with condensate-forming groups induced for 60 min and without condensates group induced for 40 min. Prior to probe addition, cells were diluted to a density corresponding to OD₆₀₀ = 1 in HBSS buffer. FDP and FDG probes were then added to the buffer at a final concentration of 100 μM and incubated with cells for 30 min at room temperature. Following incubation, cells were imaged using a STELLARIS 8 Falcon confocal microscope (Leica) using the same laser and detector setup as described above, and fluorescence signals were quantified using ImageJ (NIH).

Inter-cellular communications with DmpR circuit

To evaluate whether the catalytic activities of condensates can mediate intercellular signaling, the BL21 (DE3) competent *E. coli* was transformed with RLP_{SKGP} plasmid and the DH5 α competent *E. coli* (NEB) was transformed with the DmpR gene circuit. A single colony of the RLP_{SKGP}-harboring BL21 (DE3) cells and a single colony of the DmpR-harboring DH5 α cells were picked and each grown separately in 2 mL 2x YT liquid medium supplied with 4 % glucose and appropriate antibiotic at 37 °C overnight. The cell culture was back diluted at a ratio of 1:100 with fresh 2x YT liquid medium with appropriate antibiotic. After a 1.5-hour incubation, the BL21 (DE3) cells carrying the RLP_{SKGP} plasmid were either induced or not induced with 0.5 mM IPTG. At 3 h post induction for BL21 cell culture (4.5 h post dilution for DH5 α cell culture), each cell culture was measured for OD₆₀₀ and adjusted to the same cellular density using M9 Minimal Salts liquid medium (Sigma-Aldrich) supplied with 0.4 % (w/v) sodium acetate, 0.01 % (w/v) thiamine, 2 mM MgSO₄ and 0.1 mM CaCl₂. Each sample was prepared by mixing two cell populations at a volume ratio of 1:2, 1:1, 2:1 with the same total volume. EGFP fluorescence, absorbances at 600 nm and 410 nm were monitored with a 5-min kinetic interval at 37°C for 500 cycles on a Tecan Spark plate reader.

To further confirm the results obtained from plate reader-based monitoring of fluorescence change, we employed confocal microscopy to analyze the fluorescence from cells. Experimentally, we co-cultured BL21 (DE3) competent *E. coli* containing RLP_{SKGP} plasmid with or without condensates with DH5 α competent *E. coli* (NEB) transformed with the DmpR gene circuit, in the M9 medium with pNPP (4 mM). The co-culturing process followed the aforementioned protocol and the initial population ratio between two types of cells was 1:1. After co-culturing, the cells were diluted and subjected to confocal microscopy for further characterization. The fluorescence of EGFP was detected and the images were analyzed with ImageJ (NIH).

QUANTIFICATION AND STATISTICAL ANALYSIS

Statistical analyses were conducted using GraphPad Prism (version 8.0.1). Data are presented as mean $\pm$ SD, or as mean $\pm$ SE which indicated in the figure legends. The sample size (N) for each experiment is specified in the corresponding figure legend. Statistical significance between two groups was assessed using unpaired two-tailed t-tests, and exact *p*-values are reported in the corresponding figures. Additionally, statistical significance in [Figures 3E–3G](#) was analyzed by one-way ANOVA followed by Tukey's honestly significant difference (HSD) test.